

NETWORKING AND ROUTING PROTOCOL

ET 8315

Wednesday, April 2, 2025

Mr. Cuthbert John Karawa

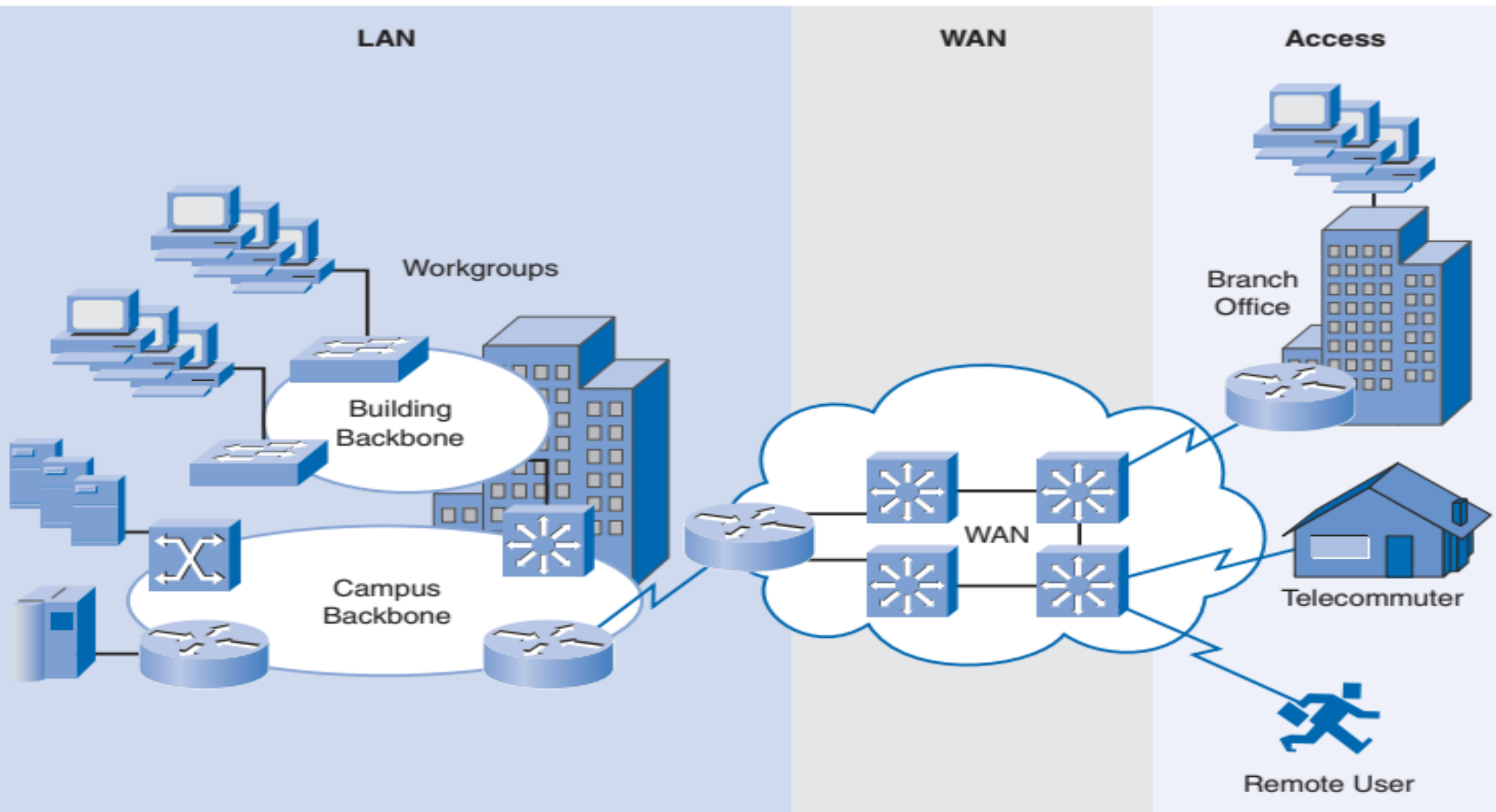
Lecture 1

WIDE AREA NETWORK (WAN)

College of Information and Communication Technology (CoICT)

Department Of Electronics And Telecommunications Engineering.

The purpose and function of WANs



What is WAN?

- One way to categorize **networks** is to divide them into **local-area networks** (LAN) and **wide-area networks** (WAN).
- LANs typically are connected workstations, printers, and other devices within a **limited geographic area** such as a building.
- All the devices in the LAN are under the **common administration of the owner of that LAN**, such as a company or an educational institution.

What is WAN?

- **Wide-area network** (WAN) is a network that spans large geographical locations, usually to interconnect multiple Local Area Networks (LANs).
- Often one particular node on a LAN is set up to serve as a **gateway** to handle all **communication going between that LAN and other networks**.
- **Others:**
 - ✓ WAN is a network that traverses a public network or commercial carrier, using one of several **WAN technologies**.

What is WAN?

- The fundamental **difference** between a WAN and a LAN is the **scalability**, WANs can be able to connect as many computers and networks as possible.
- The most popular example of WAN is the **internet**.

What is WAN?

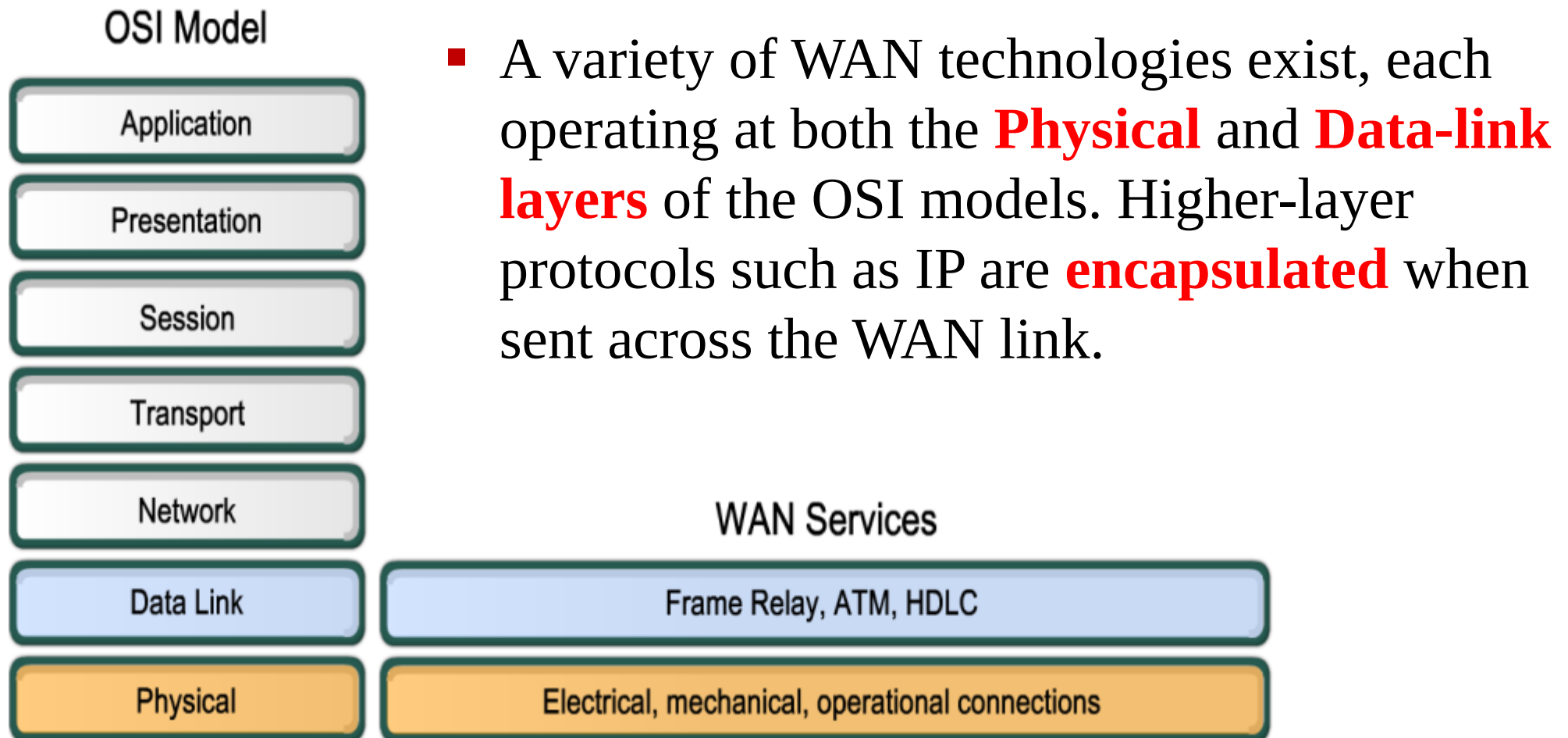
- **WANs** are different from **LANs** in several ways.
 - A LAN connects computers, peripherals, and other devices in a **single building or other small geographic area**.
 - A WAN allows the **transmission of data across greater geographic distances**.
- In addition, an **enterprise must subscribe to a WAN service provider** to use WAN carrier network services.
- LANs are typically owned by the company or organization that uses them.

Introduction

- WANS use facilities provided by a service provider, or carrier, such as a **telephone or cable company, to connect the locations of an organization to each other, to locations of other organizations, to external services, and to remote users.**
- WANS generally **carry a variety of traffic types**, such as **voice, data, and video.**

Key WAN Technology Concepts

➤ WAN functions in terms of the **OSI Reference Model**.



Key WAN Technology Concepts

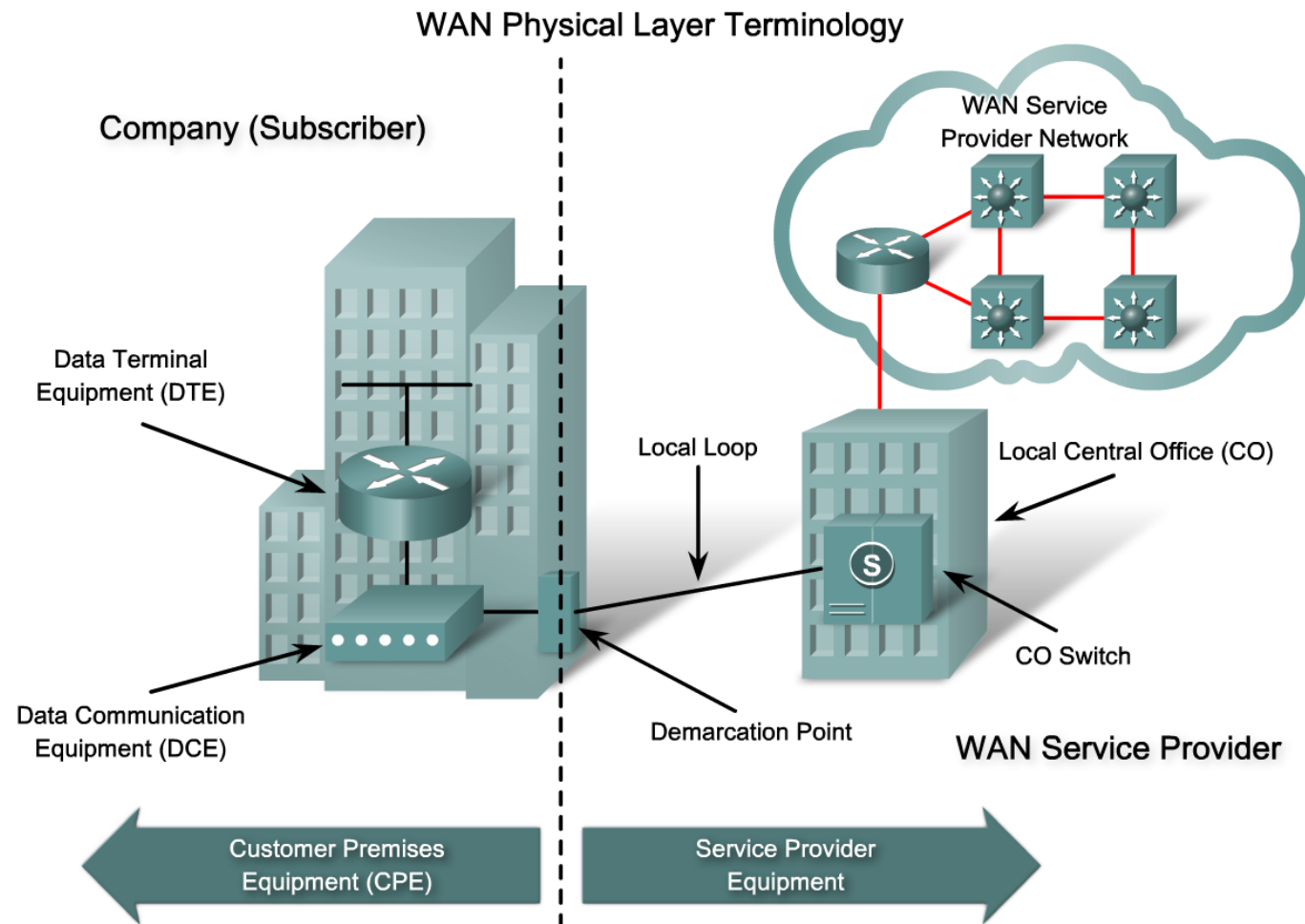
- In relation to the OSI reference model as shown in figure above, WAN operations focus primarily on **Layer 1** and **Layer 2**.
- WAN access standards typically describe both **physical layer** delivery methods and **data link layer** requirements, including **physical addressing**, **flow control**, and **encapsulation**.
- The physical layer (OSI Layer 1) protocols describe how to provide **electrical**, **mechanical**, **operational**, and **functional connections** to the services of a communications service provider.

Key WAN Technology Concepts

- The data link layer (OSI Layer 2) protocols define how data is **encapsulated** for transmission toward a **remote location** and **the mechanisms for transferring** the resulting frames.
- A variety of technologies are used, such as **Frame Relay** and **Asynchronous Transfer Mode** (ATM).
- Some of these protocols use the same basic **framing mechanism**, **High-Level Data Link Control** (HDLC), an ISO standard.

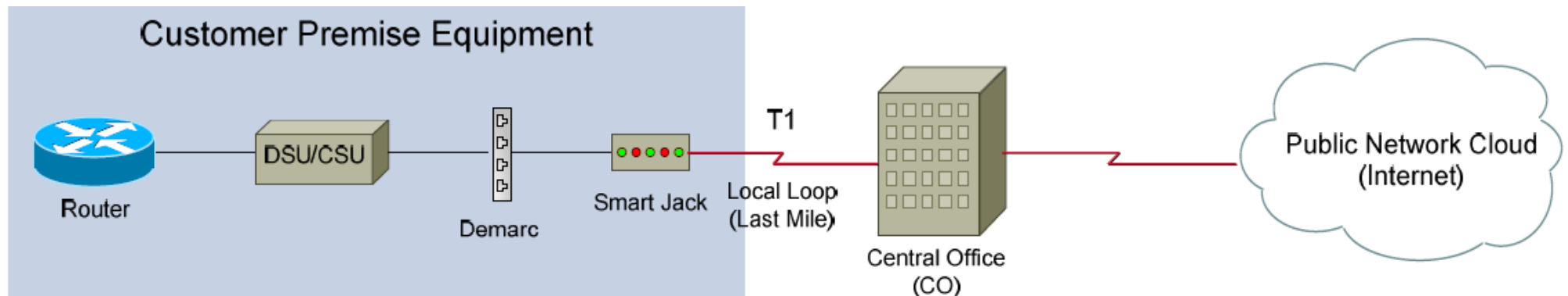
Key WAN Technology Concepts

- WAN physical layer concepts for network and Internet communications.



Common WAN Terms

- A wide variety of hardware is used with WANs, Equipment that is housed at the subscriber is referred to as **Customer Premise Equipment (CPE)**.
- The devices and inside wiring located at the premises of the subscriber, **connected with a telecommunication channel** of a carrier. The subscriber either **owns** or **leases** the CPE

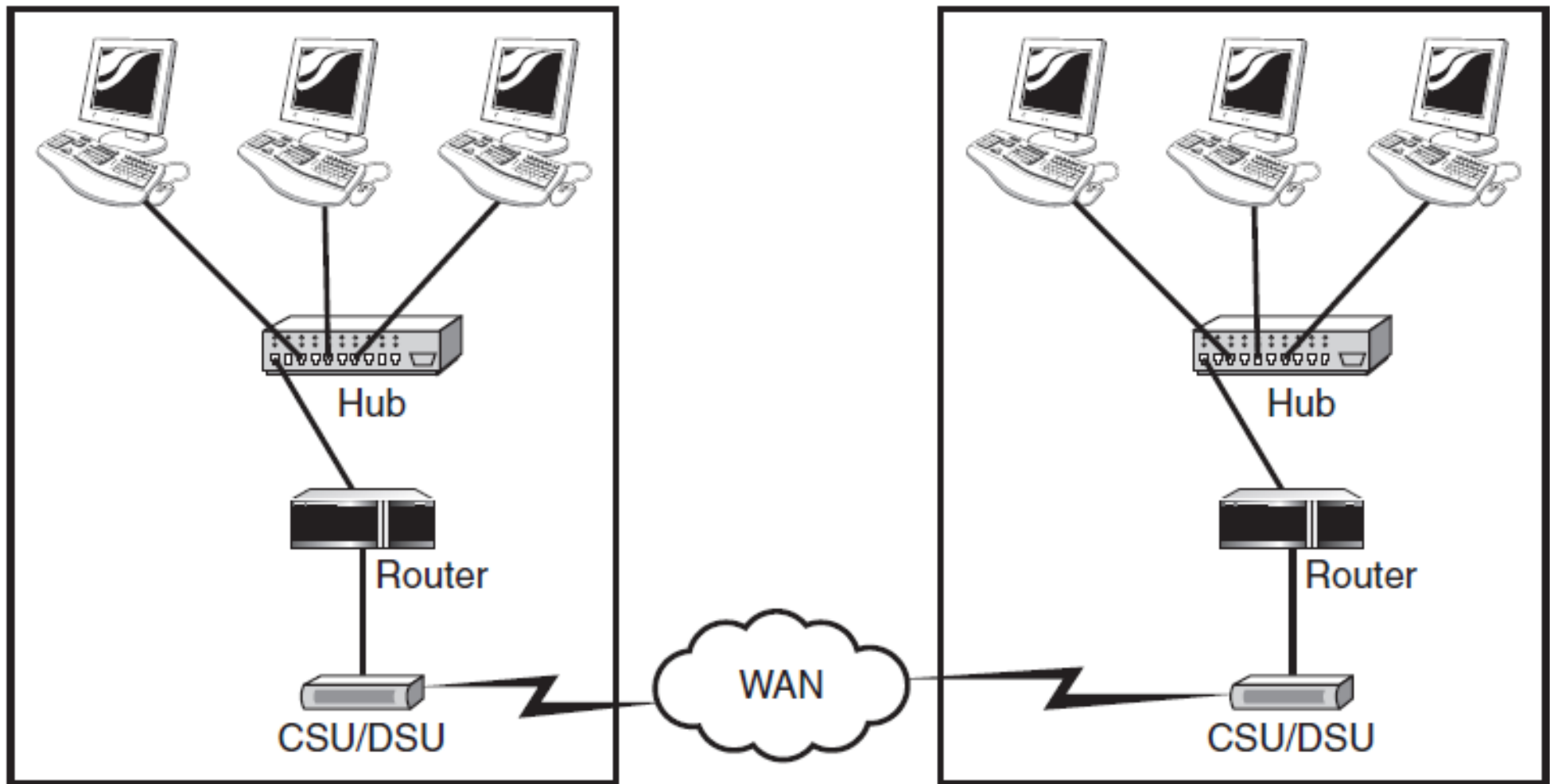


Common WAN Terms

- A **CSU/DSU (Channelized Service Unit/Data Service Unit)** provides the **clocking** and **channelization** for T1 or T3 technology. The CSU/DSU converts the signal for use on an Ethernet (or other LAN technology) network.
- If a WAN technology other than a T1 line is used, a different device will be required. Examples include (but are no limited to):
 - ✓ ISDN – a **terminal adapter**.
 - ✓ Dialup – a **modem**.
- Many routers have CSU/DSU functionality

Common WAN Terms

➤ CSU/DSU



Common WAN Terms

- The **Demarc** (short for **demarcation**) refers to the point of last responsibility for the **service provider**. All equipment on the Customer Premises side of the Demarc is the customer's responsibility to maintain. The Demarc is not **always physically labeled or identifiable**. Occasionally, a **two-port or four port patch-panel** will be used as a physical Demarc.

Common WAN Terms

- The **Smart Jack** physically terminates the T1 line. If there is a connectivity issue, the provider will perform a **ping test to the smart jack**. If communication to the smart jack is successful, the provider will assume the issue resides on the customer's side of responsibility. The smart jack is often locked in a glass enclosure, and labeled with the T1's circuit number.
- The **Local Loop** (or **Last Mile**) refers to the physical line connecting from the Customer Premises to the provider's nearest **Central Office (CO)**.

Common WAN Terms

- **Data Communications Equipment (DCE)**: Also called data circuit-terminating equipment, the DCE consists of devices that put data on the local loop. The DCE primarily provides an interface to connect subscribers to a communication link on the WAN cloud.
- **Data Terminal Equipment (DTE)**: The customer devices that pass the data from a customer network or host computer for transmission over the WAN. The DTE connects to the local loop through the **DCE**.

Common WAN Terms

- **Data Communications Equipment (DCE)**: Also called data circuit-terminating equipment, the DCE consists of devices that put data on the local loop. The DCE primarily provides an interface to connect subscribers to a communication link on the WAN cloud.
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WAN Encapsulation

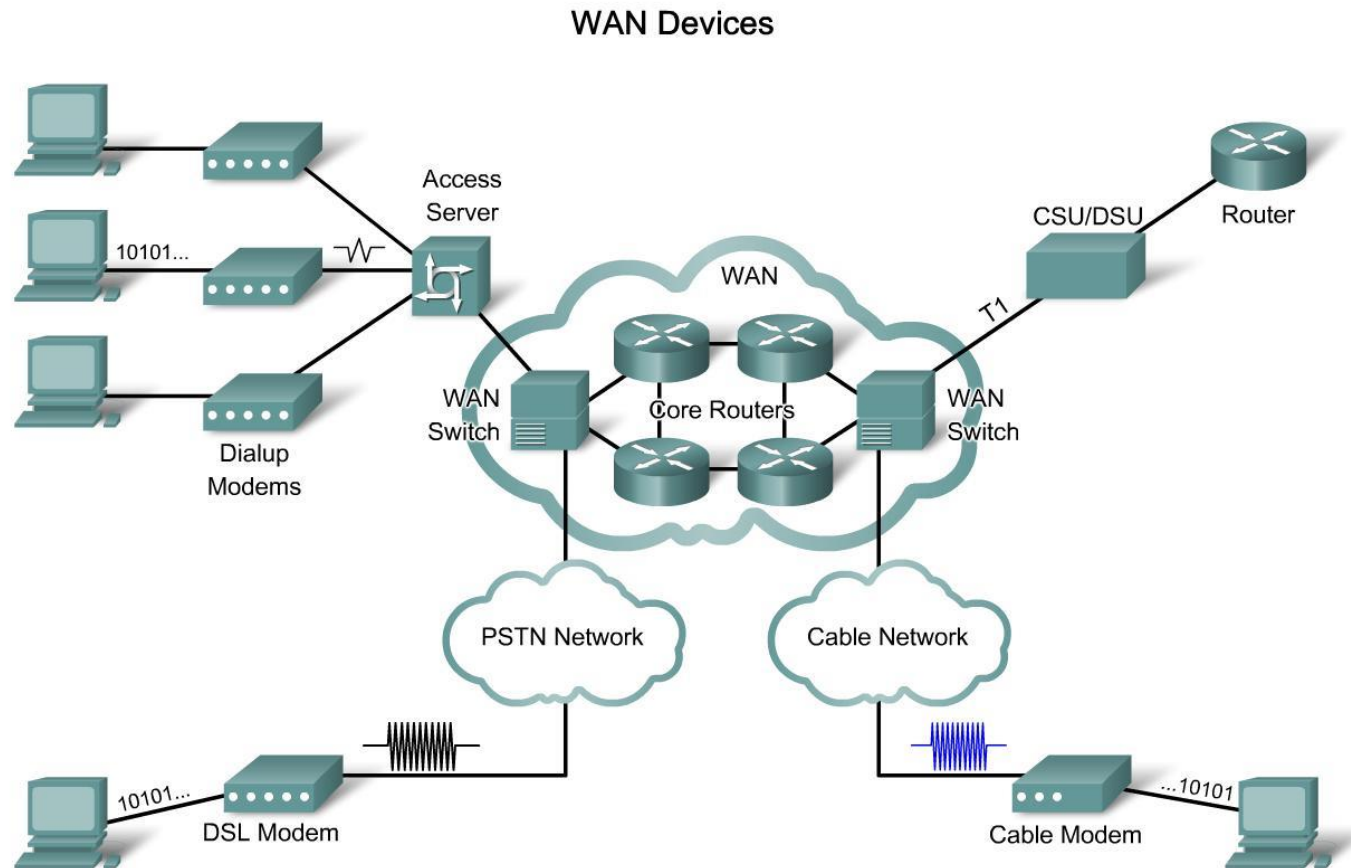
- Recall that WAN technologies operate at both **Physical** and **Data-link layers** of the OSI models, and that higher-layer protocols such as IP are **encapsulated** when sent across the WAN link.
- A WAN is usually terminated on a Cisco device's serial interface (Or any other Vendor). Serial interfaces support a wide variety of **WAN encapsulation types**, which must be manually specified.

WAN Encapsulation

- By default, a serial interface will utilize **HDLC** for encapsulation. Other supported encapsulation types include:
 - ✓ SDLC
 - ✓ Point-to-Point Protocol (PPP)
 - ✓ LAPB
 - ✓ Frame-Relay
 - ✓ X.25
 - ✓ ATM
- Regardless of the WAN encapsulation used, it must **identical** on both sides of a point-to-point link.

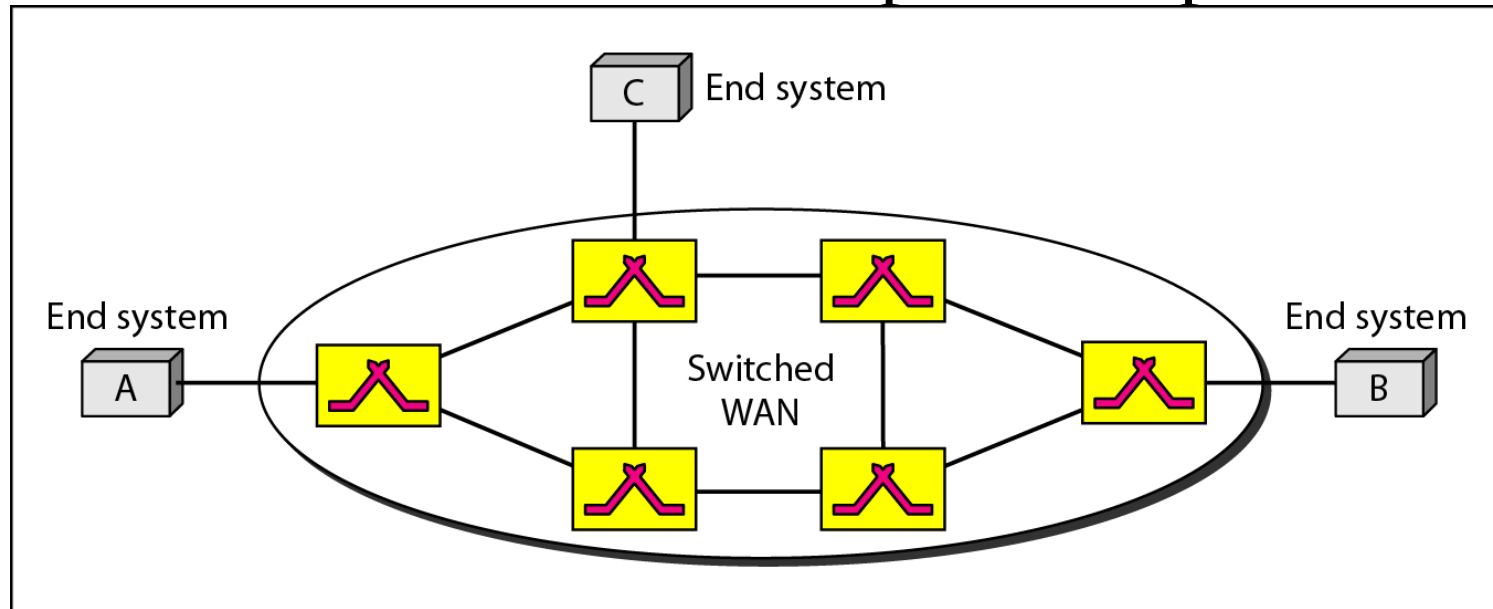
Key WAN Technology Concepts

- WAN physical layer concepts for network and Internet communications.

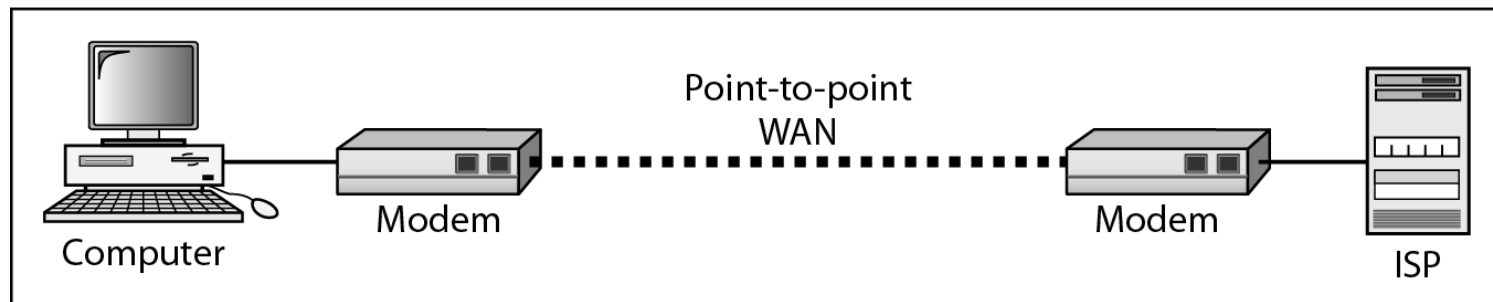


Key WAN Technology Concepts

➤ WANs: a switched WAN and a point-to-point WAN



a. Switched WAN

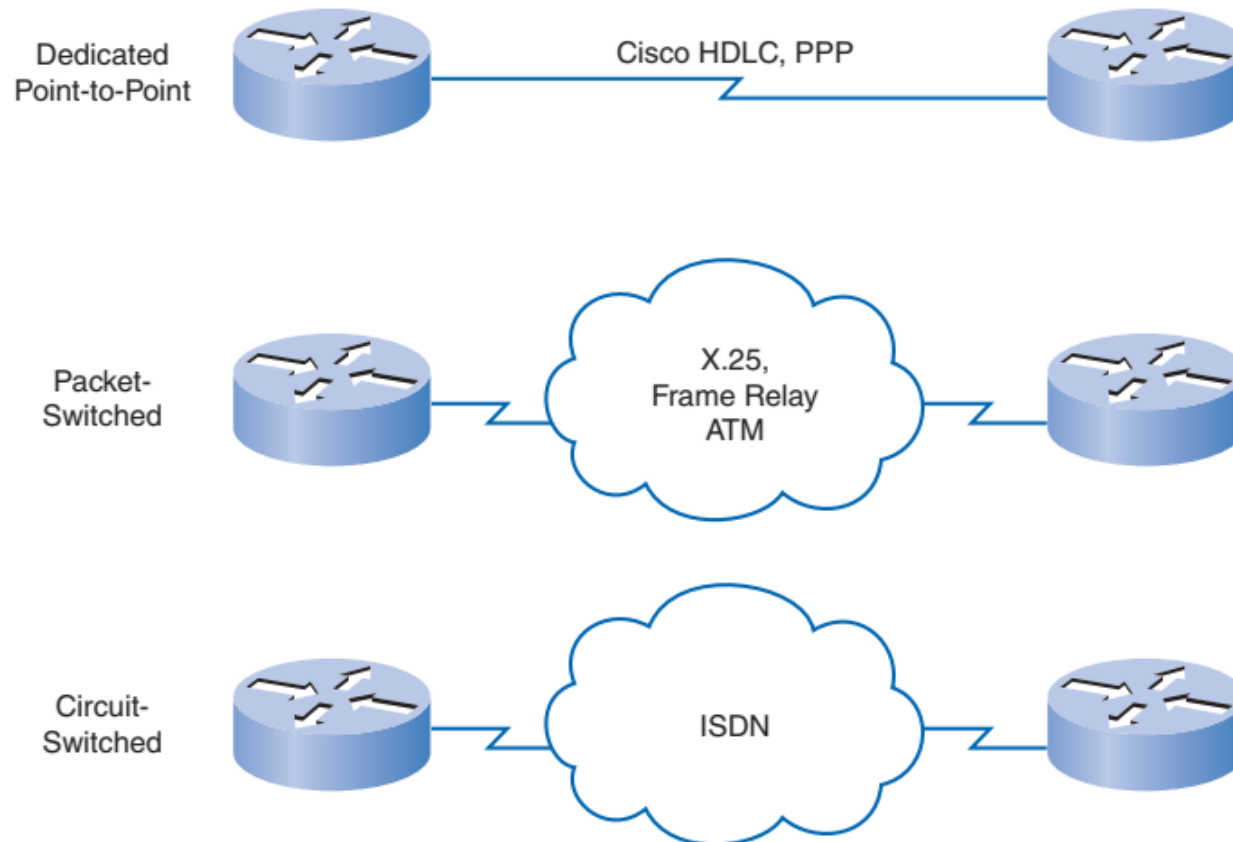


b. Point-to-point WAN

WAN Data Link Layer Concepts

- WANs require **data link layer protocols** to establish the link across the communication line from the **sending** to the **receiving** device.
- **Data link layer** protocols define how data is **encapsulated for transmission** to remote sites and the **mechanisms for transferring** the resulting frames.
- Their variety of technologies are used, such as ISDN, Frame Relay, or ATM as shown in next slide

WAN Data Link Layer Concepts



- Many of these protocols use the **same basic framing mechanism**, HDLC, an ISO standard, or one of its subsets or variants.

WAN Data Link Layer Concepts

- ATM is different from the others, because it **uses small fixed-size cells of 53 bytes (48 bytes for data)**, unlike the other packet-switched technologies, which use variable sized packets.

Protocol	Usage
Link Access Procedure Balanced (LAPB)	X.25
Link Access Procedure D Channel (LAPD)	ISDN D Channel
Link Access Procedure Frame (LAPF)	Frame Relay
High-Level Data Link Control (HDLC)	Cisco Default
Point-to-Point Protocol (PPP)	Serial WAN Switched Connections

WAN Data Link Layer Concepts

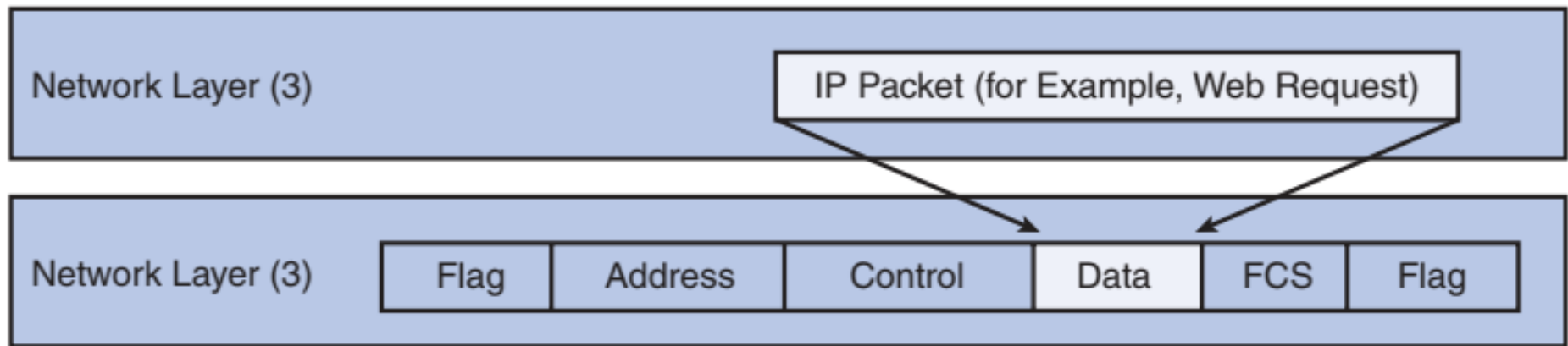
- **ISDN**(Integrated Services Digital Network) and X.25 are older data-link protocols that are less frequently used today.
- ISDN its use when provisioning a **VoIP network** using PRI(Primary Rate Interface) links (that allows traditional phone lines to carry multiple lines of voice and data).
- X.25 network is an older packet-switched network based on **Open System Interconnection(OSI)** network architecture rather than on **TCP/IP** architecture.
- X.25 is still in use in **developing countries** are used for example in aviation Industry, Automated teller machines, and by the credit card payment industry.

WAN Encapsulation

- Data from the **network layer** is passed to **the data link layer** for delivery on a physical link, which normally is **point-to-point** on a WAN connection.
- The data link layer builds a **frame** around the network layer data so that the necessary **checks** and **controls** can be applied.
- Each WAN connection type uses a **Layer 2 protocol to encapsulate a packet** while it is crossing the WAN link.
- To ensure that the correct encapsulation protocol is used, the Layer 2 encapsulation type used for **each router serial interface must be configured**.

WAN Encapsulation

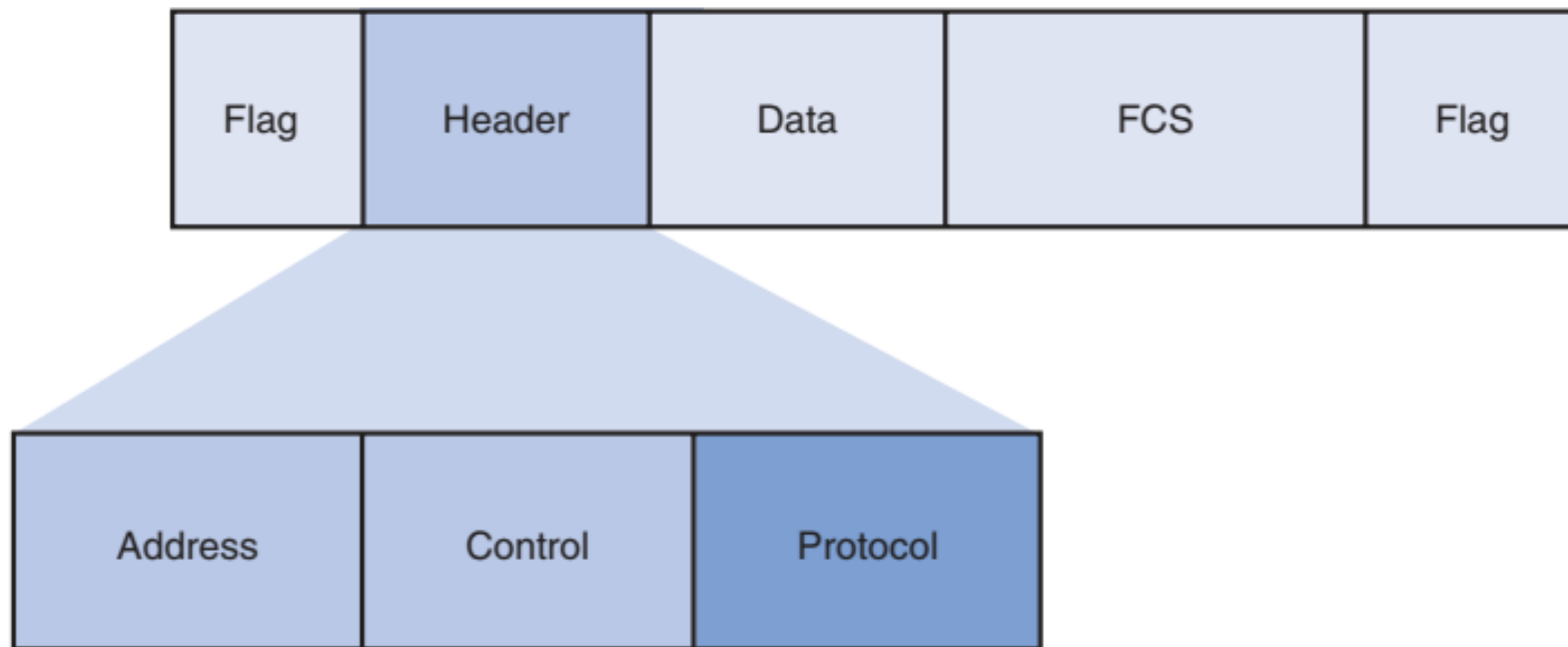
- The choice of encapsulation protocols depends on the **WAN technology** and the **equipment**.
- HDLC was first proposed in 1979; for this reason, most framing protocols that were developed afterwards are based on it.



Network data is encapsulated in an HDLC frame.

WAN Encapsulation

- If we examining the header portion of an HDLC frame, as shown in Figure below



- You can identify **common fields** used by many WAN encapsulation protocols.

WAN Encapsulation

- The frame always starts and ends with an **8-bit Flag field**. The bit pattern is 01111110.
- The Address field **is not needed** for WAN links, which are almost always point-to-point. The Address field is still present and may be **1 or 2 bytes** long.
- The Control field is protocol-dependent, but it usually indicates whether the data is **control information** or **network layer data**. The Control field normally is **1 byte**.
- The Address and Control fields, as shown in the figure are called the **frame header**.

WAN Encapsulation

- The encapsulated data follows the **Control field**. Then a frame check sequence (FCS) uses the cyclic redundancy check (CRC) mechanism to establish a 2- or 4-byte field.
- Several types of WAN encapsulation formats exist, including subsets and proprietary versions of HDLC.
- Both PPP and the Cisco version of HDLC have an **extra field** in the header to **identify the network layer protocol** of the encapsulated data.

WAN Switching Concepts

- WAN switched networks are categorized as either
 - ✓ Circuit-switched
 - ✓ Packet-switched

WAN Switching Concepts

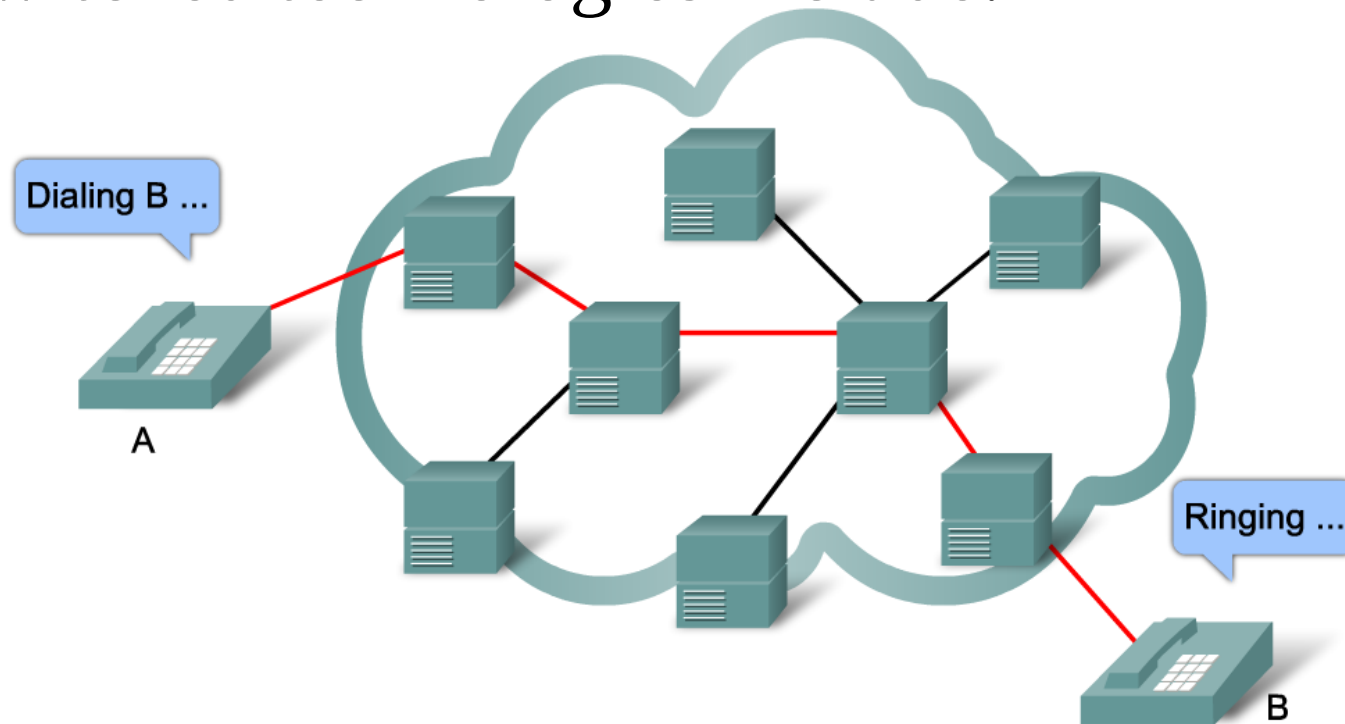
Circuit-switched

- A circuit-switched network is one that establishes a **dedicated circuit** (or channel) between **nodes** and **terminals** before the users may communicate.
- **Circuit-Switched** technologies **require call-setup to occur before** information can be transferred.
- The session is usually torn down once data transfer is complete (this is identified as an **On-Demand Circuit**).

WAN Switching Concepts

➤ Circuit switched lines are generally **low-speed** compared to point-to-point lines. Ordinary **voice phone service** uses circuit switching. Examples of circuit-switched technologies include:

- ✓ Dial-up
- ✓ ISDN



WAN Switching Concepts

- The internal path taken by the circuit between exchanges is shared by a number of conversations.
- **Time-division multiplexing (TDM)** gives each conversation a share of the connection in turn. TDM ensures that a fixed-capacity connection is made available to the subscriber.
- If the circuit carries computer data, the usage of this fixed capacity may be inefficient.

WAN Switching Concepts

- **PSTN** and **ISDN** are two types of **circuit-switching** technology that may be used to implement a WAN in an enterprise setting.

WAN Switching Concepts

Packet Switching

- In contrast to circuit switching, packet switching **splits traffic data into packets** that are **routed over a shared network**.
- Packet-switching networks **do not require a circuit** to be established, and they **allow many pairs of nodes to communicate over the same channel**.
- Packet-Switched technologies **share a common infrastructure** between all the provider's subscribers.

WAN Switching Concepts

- Thus, **bandwidth is not guaranteed**, but is instead allocated on a **best effort basis**.
- Packet-switched technologies are ill-suited for applications that require consistent bandwidth, but are considerably **less expensive than dedicated point-to-point lines**.
- Examples of packet-switched technologies include:
 - ✓ Frame-Relay
 - ✓ X25

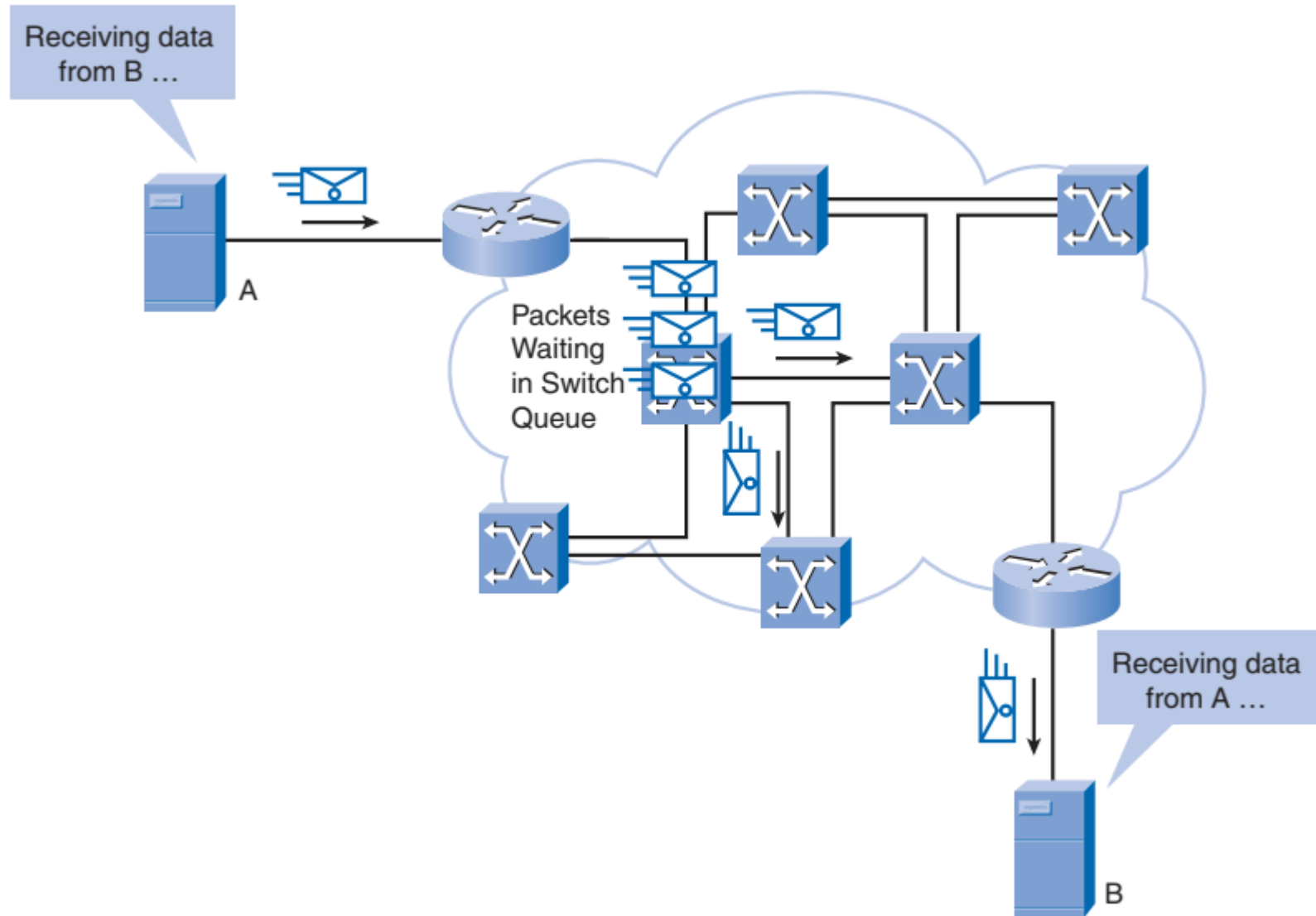
WAN Switching Concepts

- The switches in a packet-switched network **determine which link the packet** must be sent on next from the addressing information in each packet.
- There are two approaches to this link determination:
 - ✓ **Connectionless systems**, such as the Internet, carry full addressing information in each packet. Each **switch must evaluate the address** to determine where to send the packet.
 - ✓ **Connection-oriented systems** predetermine a packet's route, and each packet only has to carry an identifier. The **switch determines the onward route** by looking up the identifier in tables held in memory.

WAN Switching Concepts

- Because the internal links between the switches are shared between many users, **the costs of packet switching are lower than those of circuit switching.**
- **Delays** (latency) and **variability of delay** (jitter) are **greater** in packet-switched networks than in circuit-switched networks. This is because the **links are shared**, and **packets must be entirely received at one switch before moving to the next.**
- Despite the **latency** and **jitter** inherent in shared networks, modern technology allows satisfactory transport of voice and even video communications on these networks.

WAN Switching Concepts



WAN Switching Concepts

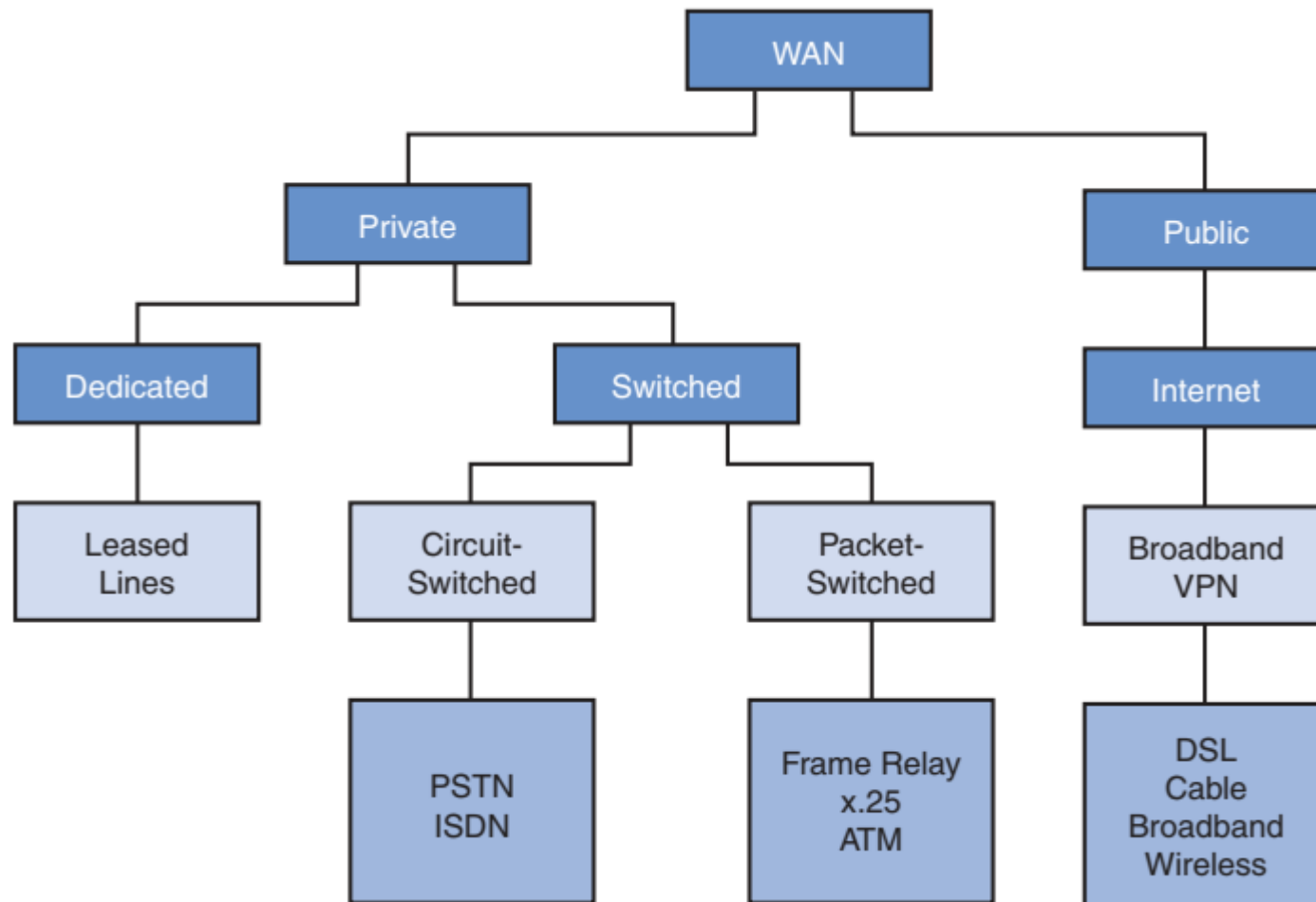
- From the figure above **Server A** is sending data to **Server B** the packets may not necessarily always take the same path to reach the destination. Each packet may take **different routes** to reach Server B.
- To connect to a packet-switched network, a subscriber needs a **local loop to the nearest location** where the provider makes the service available. This is called the **service's point of presence (POP)**.

WAN Connection Options

- WAN solutions are currently available. They differ in **technology**, **speed**, and **cost**.
- Familiarity with these technologies is an important part of **network design** and **evaluation**.
- There various WAN connection options, including **private dedicated links**, **private switched links**, and **public connection options** using the Internet.

WAN Connection Options

➤ WAN Link Connection Options



WAN Connection Options

➤ **Private WAN connection options** it include both dedicated and switched communication link options:

✓ **Dedicated communication links:**

- When permanent dedicated connections are required, point-to-point lines are used with various capacities that are limited only by the underlying physical facilities and the willingness of users to pay for these dedicated lines.

WAN Connection Options

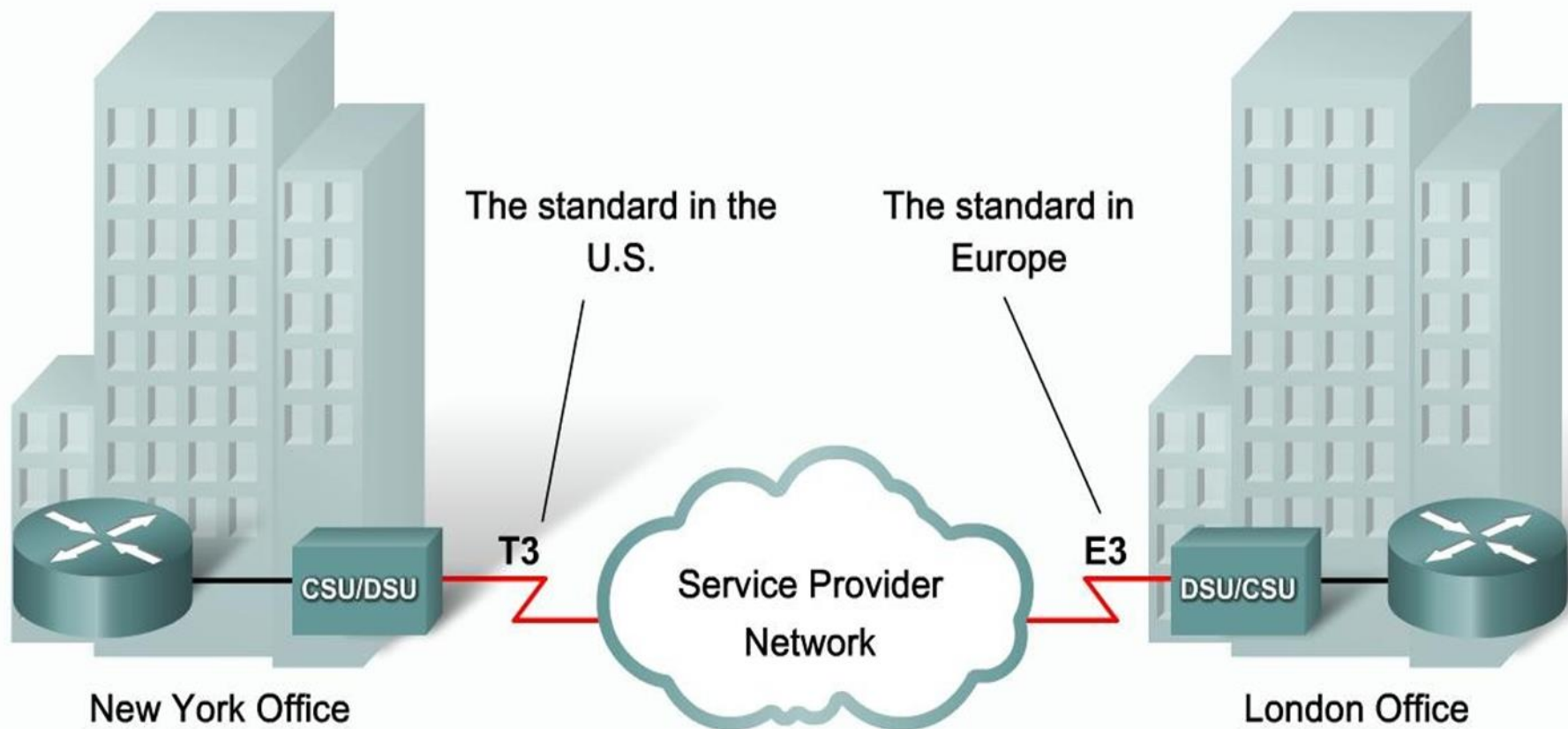
- A point-to-point link provides a preestablished WAN communications path from **the customer premises through the provider network to a remote destination**.
- Point-to-point lines usually are leased from a carrier and are also called **leased lines**.

➤ Leased lines are available in different **capacities**. They generally are priced based on the **bandwidth** required and the **distance** between the two connected points.

WAN Connection Options

- Leased line services to provide a WAN connection

Leased Lines



WAN Connection Options

- **Public WAN connection options** this use the global Internet infrastructure.
- The Internet was not a viable networking option for many businesses because of the significant security risks and lack of adequate performance guarantees in an end-to-end Internet connection.
- With the development of VPN technology, however, the Internet is now an **inexpensive and secure option** for connecting to teleworkers and remote offices where performance guarantees are not critical.

WAN Connection Options

- Internet WAN connection links are through **broadband services** such as **Digital Subscriber Line (DSL)**, **cable modem**, and **broadband wireless**, and they are combined with **VPN technology** to provide privacy across the Internet.

WAN Connection Options

- **Point-to-Point technologies** (often called **dedicated** or **leased** lines) are usually the most expensive form of WAN technology.
- Point-to-Point technologies are **leased** from a service provider, and provide **guaranteed bandwidth** from location to another (hence point-to-point).
- Cost is determined by the **distance** of the connection, and the amount of **bandwidth** allocated.

WAN Connection Options

- Generally, point-to-point links require **no call-setup**, and the connection is usually **always on**. Examples of point-to-point technologies include:
 - ✓ T1 lines
 - ✓ T3 lines
- They have been the traditional connection of choice but have a **number of disadvantages**.
- Leased lines have a fixed capacity, WAN traffic is often variable, leaving some of the capacity unused.

WAN Connection Options

- In addition, each endpoint **needs a separate physical interface on the router**, which increases equipment costs. Any changes to the leased line generally require a site visit by the carrier.

WAN Connection Options

Circuit-Switched Connection Options

- Circuit-switched networks establish a **dedicated connection** for voice or data between a sender and a receiver.
- Before any communications can begin, it is necessary to establish the connection through the service provider's network.
- Circuit-Switched technologies require **call-setup** to occur before information can be transferred.
- The session is usually torn down once data transfer is **complete** (this is identified as an **On-Demand Circuit**).

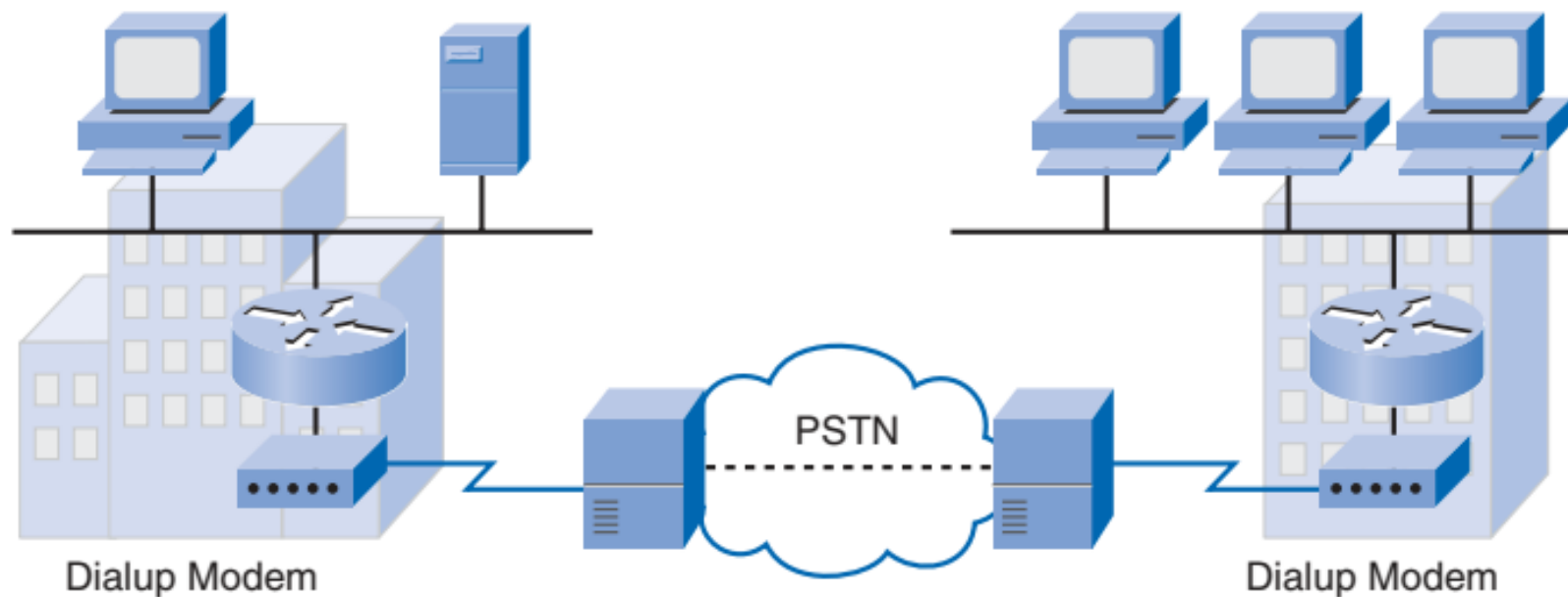
WAN Connection Options

- Circuit switched lines are generally low-speed compared to point-to-point lines.
- Examples of circuit-switched technologies include:
 - ✓ Dial-up
 - ✓ ISDN

WAN Connection Options

Analog Dialup:

- When **intermittent, low-volume data transfers** are needed, modems and analog dialed telephone lines provide low capacity and dedicated switched connections.



WAN Connection Options

- The above figure shows WAN Built with an Intermittent Connection Using a Modem and the Voice Telephone Network.
- Traditional telephony uses a **copper cable**, called the **local loop**, to connect the telephone handset in the subscriber premises to the CO.
- The signal on the local loop during a call is a continuously varying electronic signal that is a translation of the subscriber analog voice signal.

WAN Connection Options

- Traditional local loops can **transport binary computer data** through the voice telephone network using a modem.
- The modem **modulates** the binary data into an analog signal at the source and **demodulates** the analog signal into binary data at the destination.
- The physical characteristics of the local loop and its **connection to the PSTN** limit the signal's rate to less than **56 kbps**.

WAN Connection Options

- Using automatic dialup at night or on **weekends for large file transfers and data backup** can take advantage of lower **off-peak tariffs** (line charges).
- Tariffs are based on the **distance** between the endpoints, time of day, and the call's duration.
- The **advantages** of modem and analog lines are **simplicity, availability, and low implementation cost**.
- The **disadvantages** are the **low data rates** and a relatively **long connection time**.
- The dedicated circuit has little delay or jitter for point-to-point traffic, but voice or video traffic does not operate adequately at these low bit rates.

WAN Connection Options

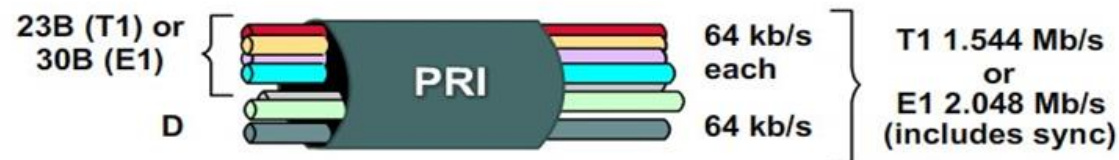
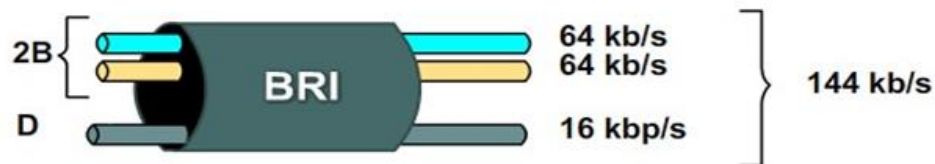
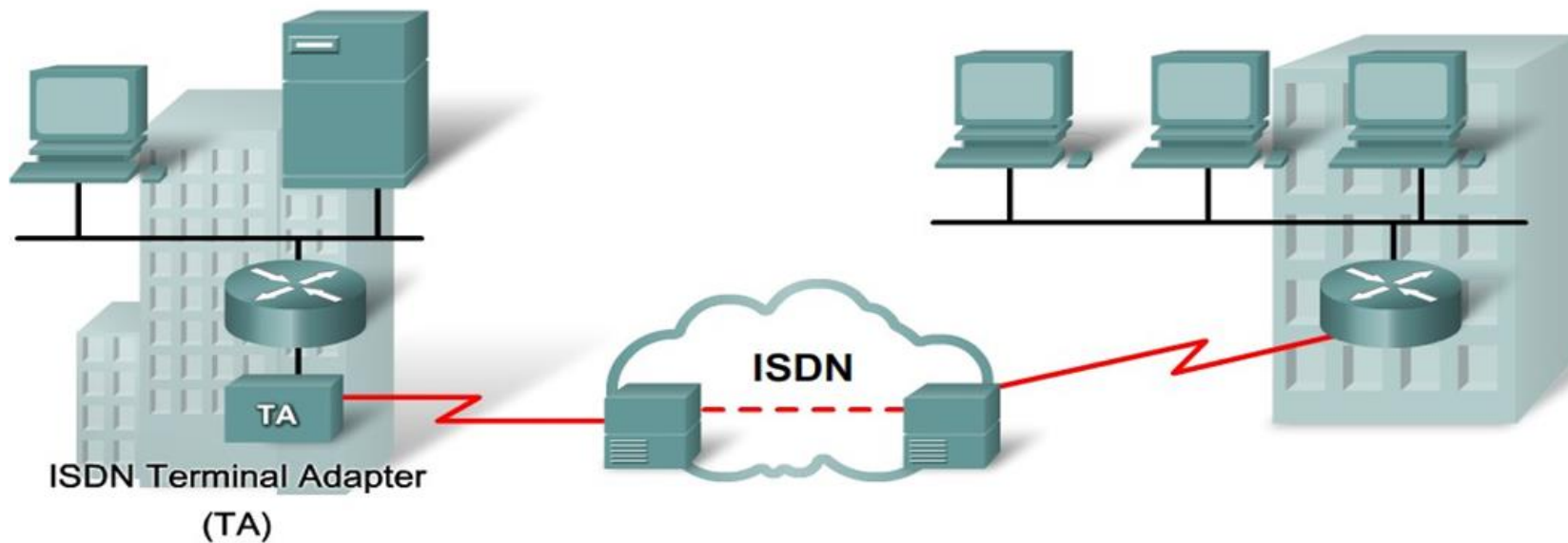
- Integrated Services Digital Network(**ISDN**) is a circuit-switching technology that enables the local loop of a PSTN to carry digital signals, resulting in higher-capacity switched connections.
- ISDN changes the internal connections of the PSTN from carrying analog signals to **time-division multiplexed** (TDM) digital signals.
- TDM allows **two or more signals or bit streams to be transferred as subchannels in one communication channel.**

WAN Connection Options

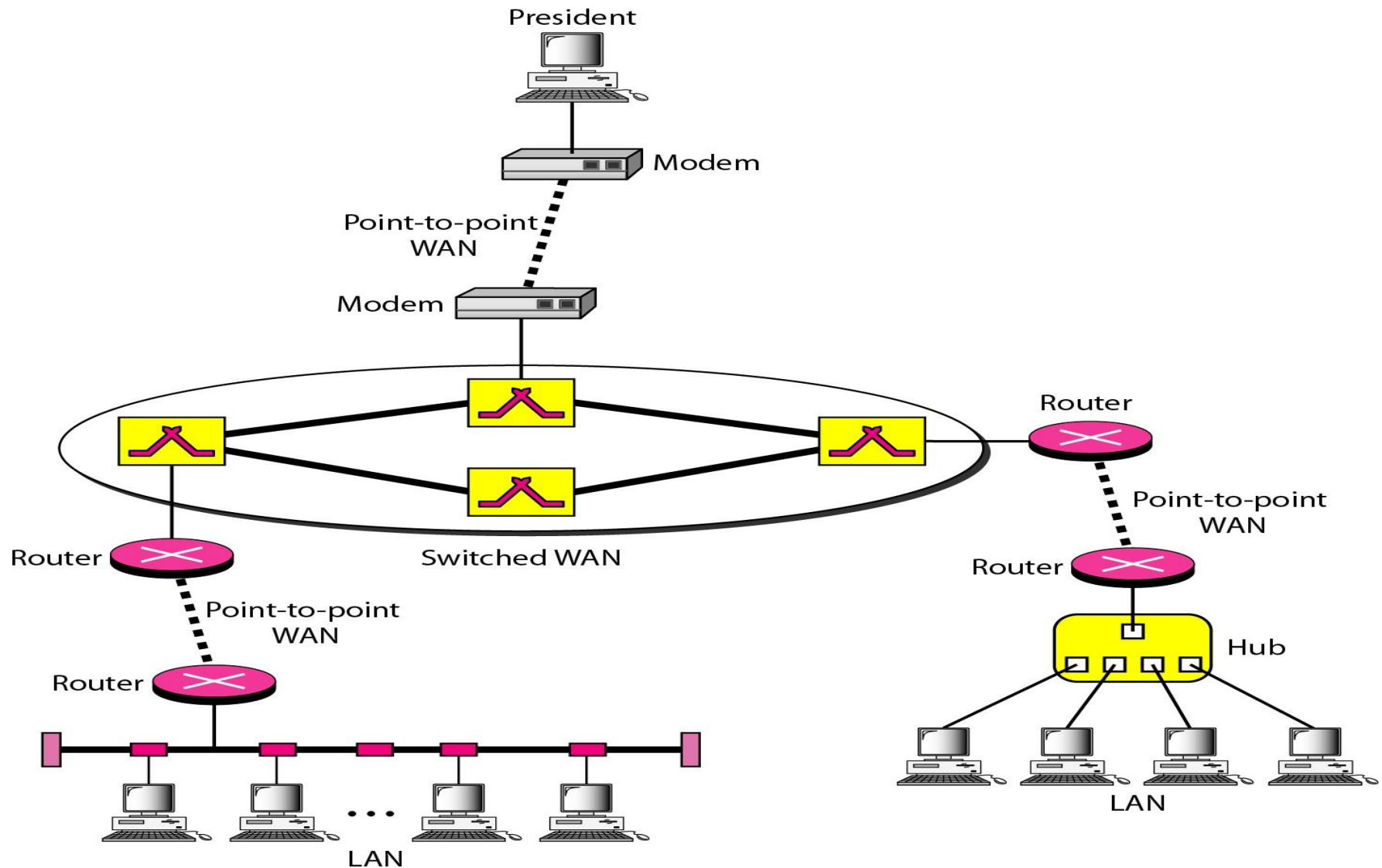
- There are two types of ISDN interfaces:
 - ✓ **Basic Rate Interface (BRI):** This is intended for the home and small enterprise and provides two 64-kbps B channels and a 16-kbps D channel.
 - ✓ **Primary Rate Interface (PRI):** this is also available for larger installations. PRI delivers 23 B channels with 64 kbps and one D channel with 64 kbps in North America, for a total bit rate of up to 1.544 Mbps.

WAN Connection Options

ISDN



Heterogeneous Network



Select the Appropriate WAN Technology

➤ Factors to consider when selecting a WAN connection

Option	Description	Advantages	Disadvantages	Sample Protocols
Leased line	Point-to-point connection between two computers' LANs.	Most secure	Expensive	PPP, HDLC, SDLC
Circuit switching	A dedicated circuit path is created between endpoints. Best example is dialup connections.	Less expensive	Call setup	PPP, ISDN

Select the Appropriate WAN Technology

➤ Factors to consider when selecting a WAN connection

Packet switching	Devices transport packets via a shared single point-to-point or point-to-multipoint link across a carrier inter network. Variable-length packets are transmitted over PVCs or SVCs.	Highly efficient use of bandwidth	Shared media across link	X.25, Frame Relay
Cell relay	Similar to packet switching, but uses fixed-length cells instead of variable-length packets. Data is divided into fixed-length cells and then transported across virtual circuits.	Best for simultaneous use of voice and data	Overhead can be considerable	ATM

Select the Appropriate WAN Technology

➤ Factors to consider when selecting a WAN connection

Internet	Connectionless packet switching using the Internet as the WAN infrastructure. Uses network addressing to deliver packets. Because of security issues, VPN technology must be used.	Least expensive, globally available	Least secure	VPN, DSL, cable modem, wireless
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SUMMARY

- Selecting the appropriate WAN technology requires considering some of the following:
 - ✓ WAN's purpose
 - ✓ Geographic scope of WAN
 - ✓ Traffic requirements
 - ✓ If WAN uses a public or private infrastructure

WAN Devices

- **Routers:** Provides WAN interface ports
- **Switches:** Provide connectivity for voice, data, and video communication
- **Modems:** Interface voice-grade services
 - ✓ CSU/DSUs that interface T1/E1 services
 - ✓ TA/NT1s (ISDN modem) that interface ISDN services
- **Communication servers**
 - ✓ Dial-in and dial-out user communication
 - ✓ Manages the dial-up
 - ✓ Log-on authentication

WAN Devices

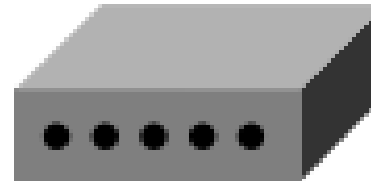
Router



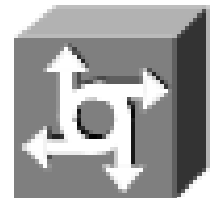
WAN
Bandwidth
Switch



Modem



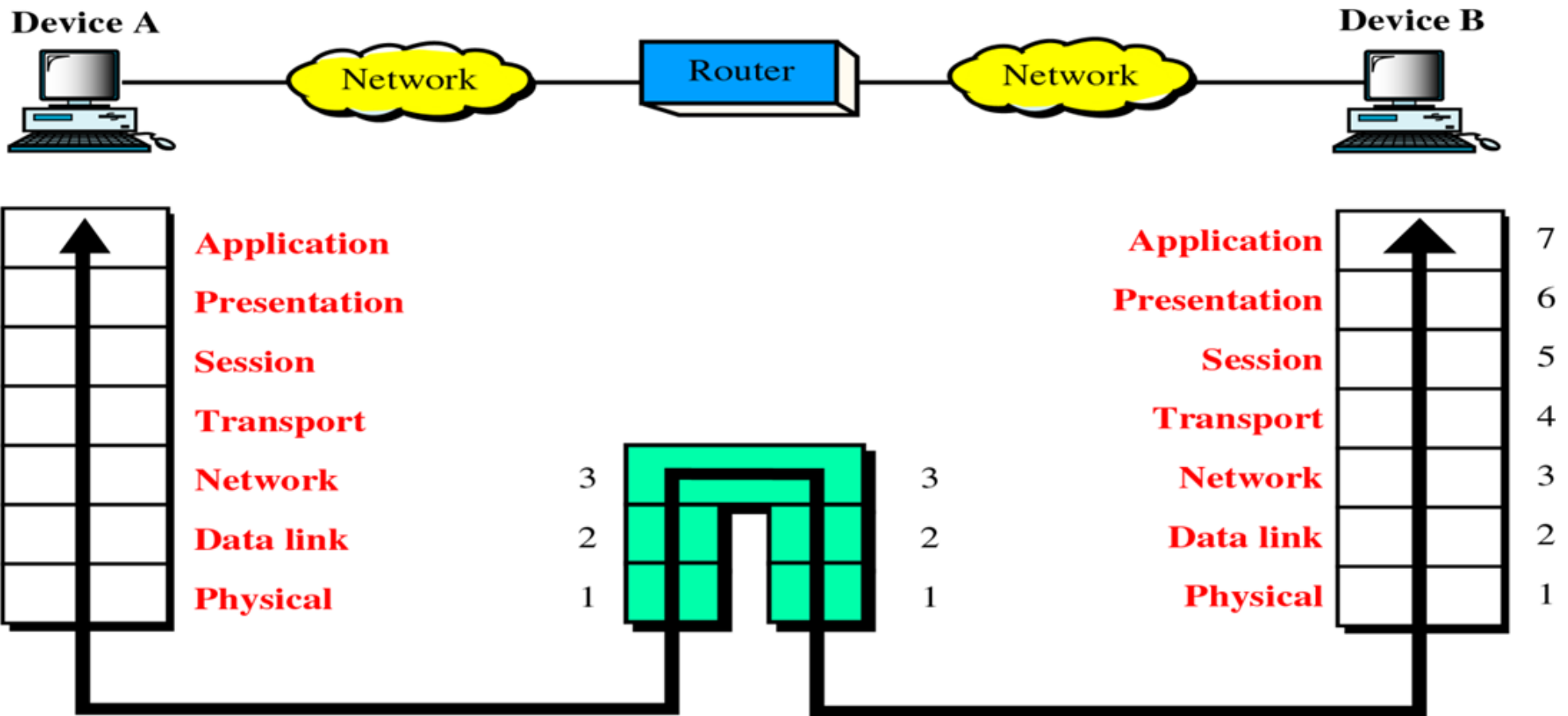
Comm.
Server



Routers

- **Routers** have access to **network layer addresses** and contain software that enables them to determine which of several possible paths between those addresses is the best for a particular transmission.
- Routers operate in the **physical, data link, and network layers** of the OSI model.

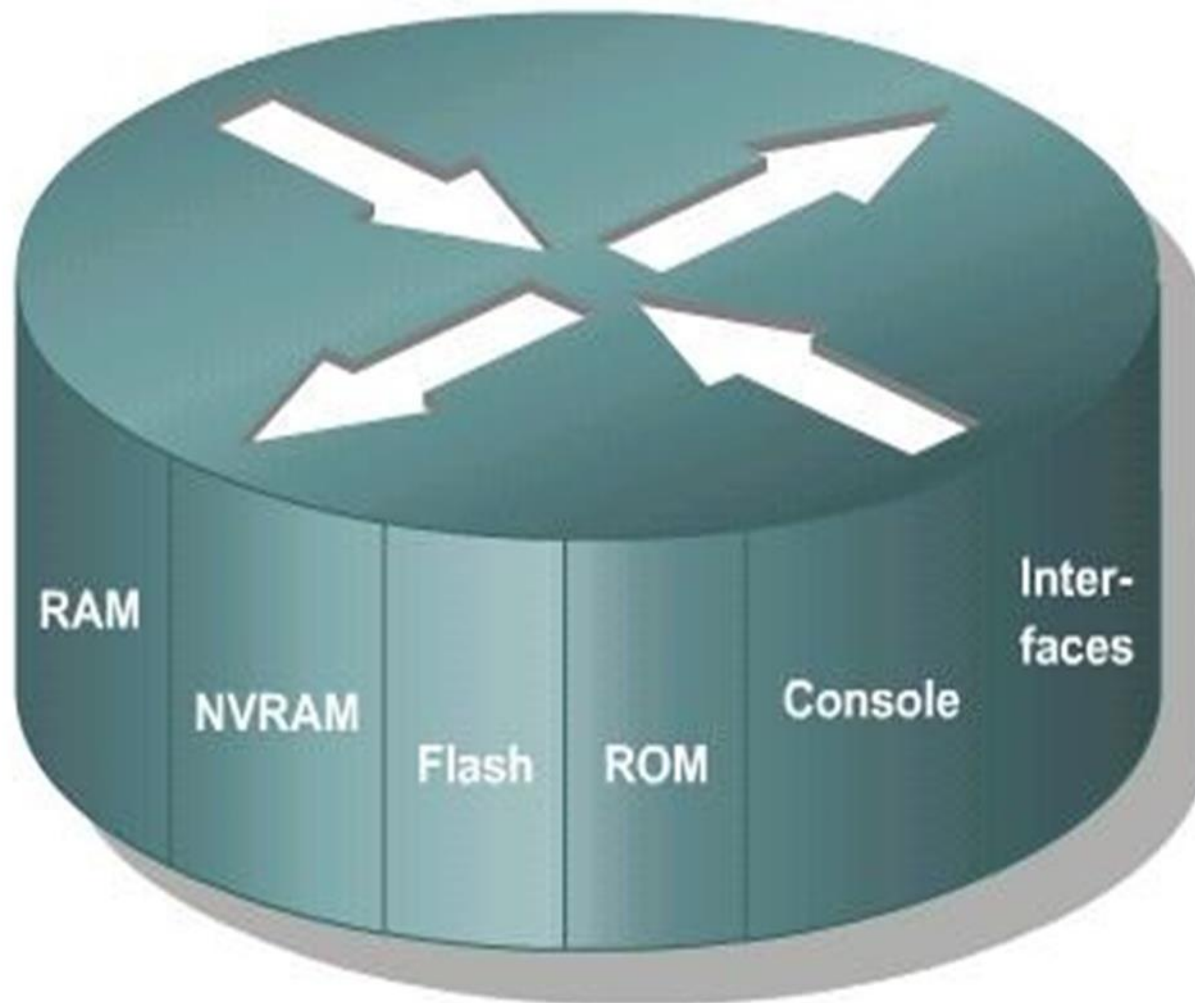
Routers



Routers Components

- Like Computers, Routers has the following four basic components:
 1. CPU
 2. Memory
 3. Interfaces
 4. Bus
- Therefore it can be referred to as a computer.
- Routers need the **Internetworking Operating Software (IOS)** to run configuration files just like a computer needs an OS to run software applications.

Routers Components



Routers Components

- RAM, or dynamic RAM (DRAM) is usually logically divided into main processor memory and shared input/output (I/O) memory
 - ✓ Stores routing tables
 - ✓ Holds ARP cache
 - ✓ Holds fast-switching cache
 - ✓ Performs packet buffering (shared I/O memory)
 - ✓ Maintains packet-hold queues
 - ✓ Provides temporary memory for the configuration file of the router while the router is powered on
 - ✓ Loses content when router is powered down
 - ✓ Can be upgraded by adding Dual In-Line Memory Modules (DIMMs)

NVRAM AND FLASH

➤ **Non-Volatile RAM (NVRAM)**

- ✓ Store backup/startup config file
- ✓ Retains contents when powered down or restarted.

➤ **Flash**

- ✓ Erasable programmable ROM (EPROM)
- ✓ Upgraded with Single In-Line Memory Modules (SIMMs) or Flash Cards (PCMCIA) without removing chips on board
- ✓ Multiple versions of IOS can be stored in the flash
- ✓ Contents remain when powered down
- ✓ In most routers the IOS is transferred to RAM during the boot process. In others the IOS may be run directly from flash

ROM and Interfaces

➤ **ROM**

- ✓ Maintains instructions for POST
- ✓ Stores bootstrap program and basic IOS.
- ✓ Software upgrades in ROM require replacing chips on the CPU
- ✓ Stores a scaled down version of the IOS

➤ **Interfaces**

- ✓ Network connections through which packets enter and exit
- ✓ Located on motherboard or on a separate module

BUS

- **Buses:** Most routers contain a system bus and a CPU bus.
- ✓ The system bus transfers packets between the **CPU** and **interfaces**.
- ✓ The CPU bus transfers instructions and data between the CPU and specified memory addresses

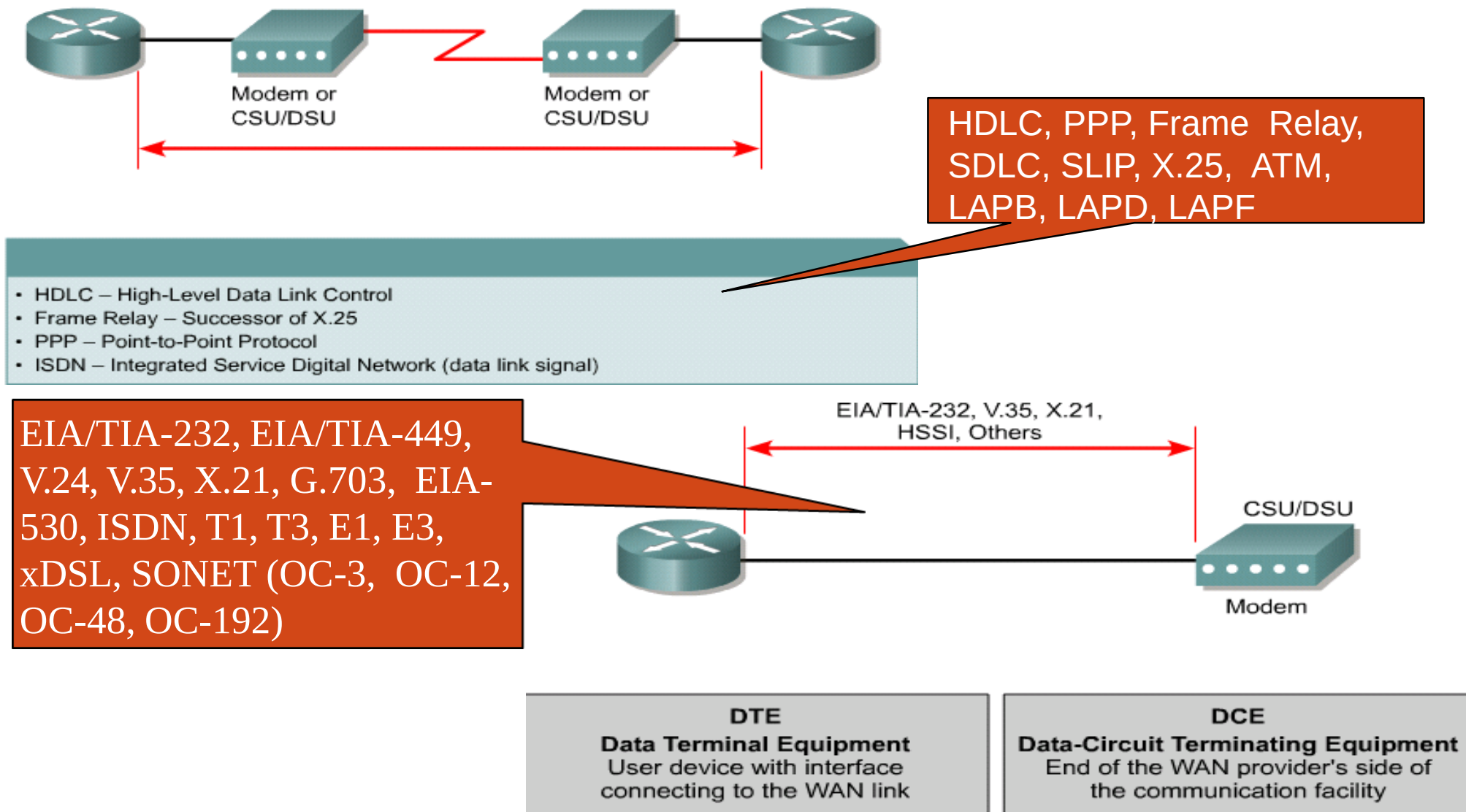
Segmentation with Routers

- Can be used to **segments LANs**, but generally used as a WAN device.
- Routers have both **LAN** and **WAN** interfaces
- Routers are the backbone devices of large intranets and of the Internet
- A correctly configured internetwork provides:
 - ✓ Consistent end-to-end addressing
 - ✓ Addresses that represent network topologies
 - ✓ Best path selection
 - ✓ Dynamic or static routing
 - ✓ Switching.

Router Role in a WAN

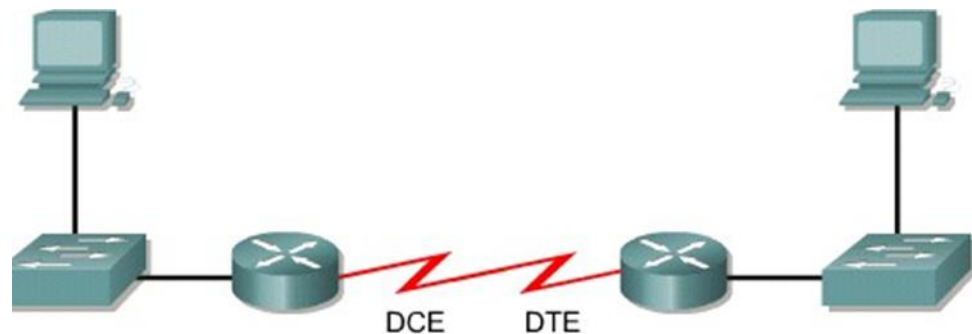
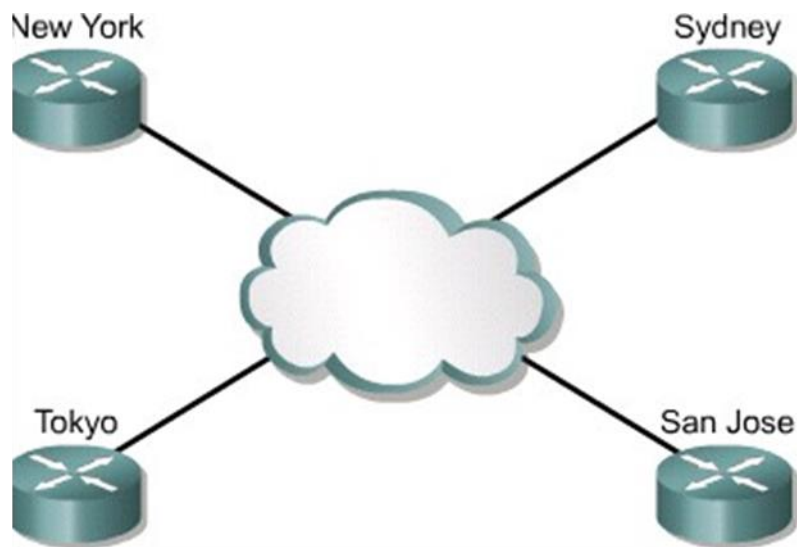
- WANs operate at the **Physical** and **Data Link** layers
- The standards and protocols used in WANs at **Layer 1 and 2** are different from those used in LANs at the same layers.
- Data link layer protocols describe how frames are carried between systems on a single data link.
- Physical layer describes the interface between the DTE and the DCE

WAN Layer 1 and 2 Protocols



LAB Setup

- Unlike the lab setup, the serial cables in the real world are not connected back to back
- In the lab, devices that make up the WAN cloud are simulated by the connection between the back-to-back DTE-DCE cables.

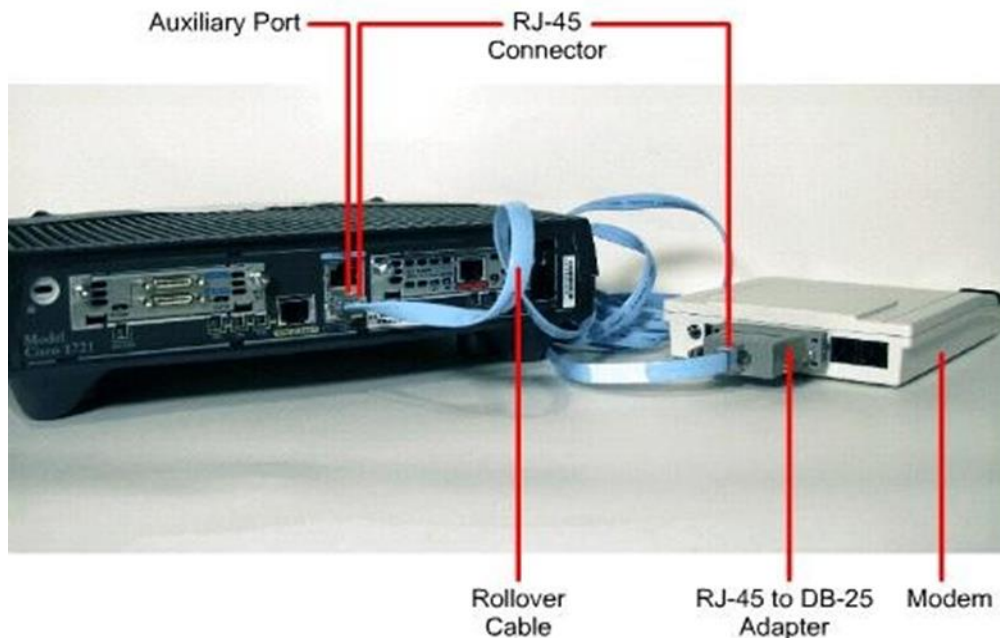


PORTS

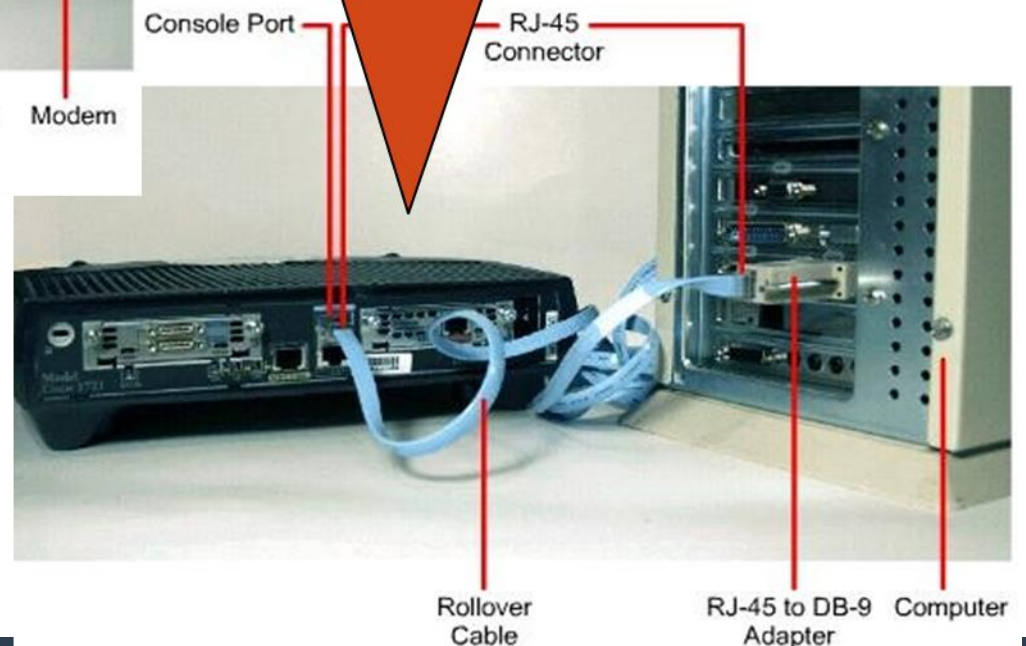
3 TYPES OF INTERFACES

1. **LAN:** Ethernet, Token Ring or FDDI
2. **WAN:** Serial, ISDN, and integrated Channel Service Unit (CSU)
3. **Management:** Console and Aux port
 - ✓ EIA-232 asynchronous serial ports connects to the computers COM port
 - ✓ Computers must use a **terminal emulation** program that provides a text-based session with the router
 - ✓ DB9-RJ45 connector and a rollover cable not designed as networking ports

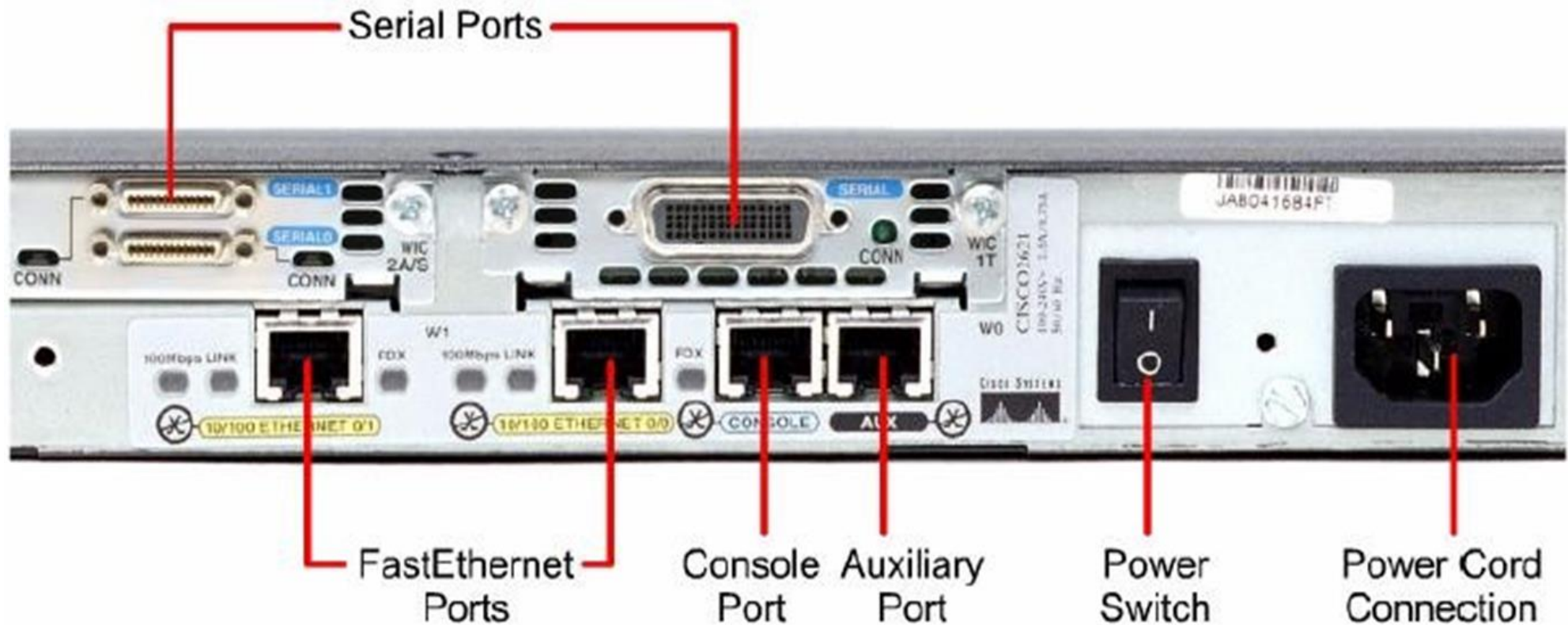
Management port connections



- Recommended for initial configuration
- Displays router startup, debugging and error messages by default
- Used for password recovery procedures



Router external connections

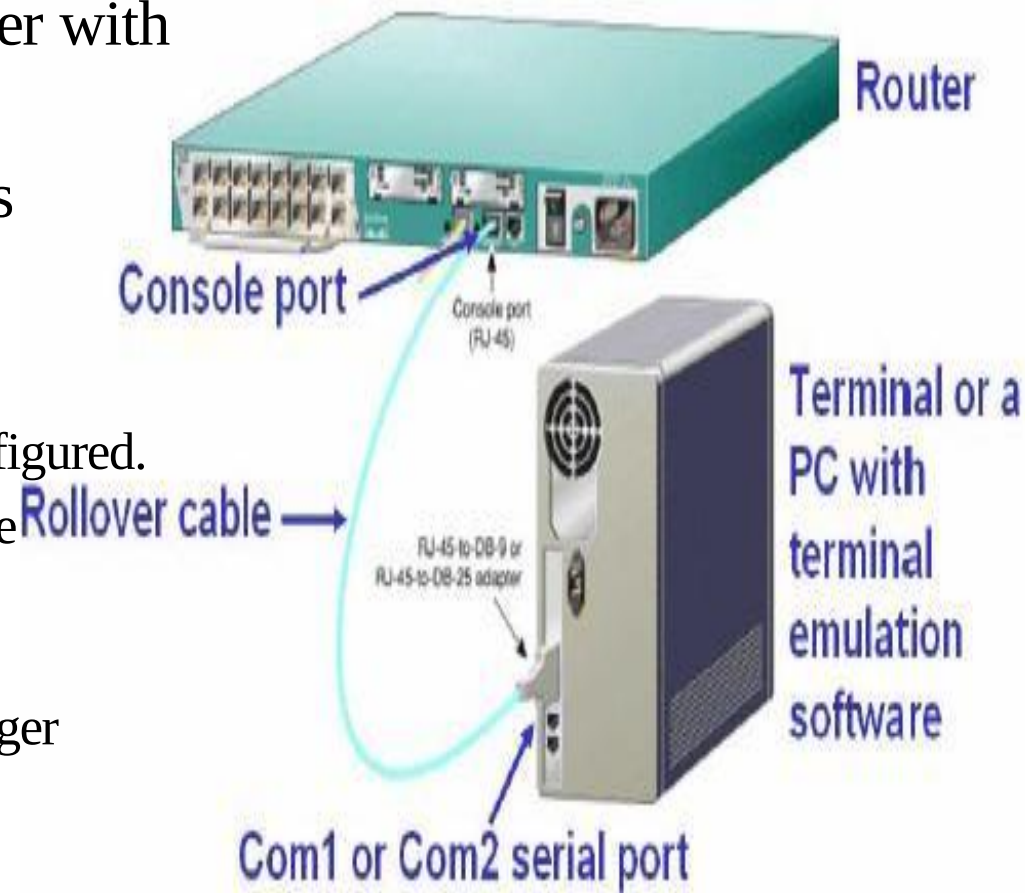


Accessing the Router

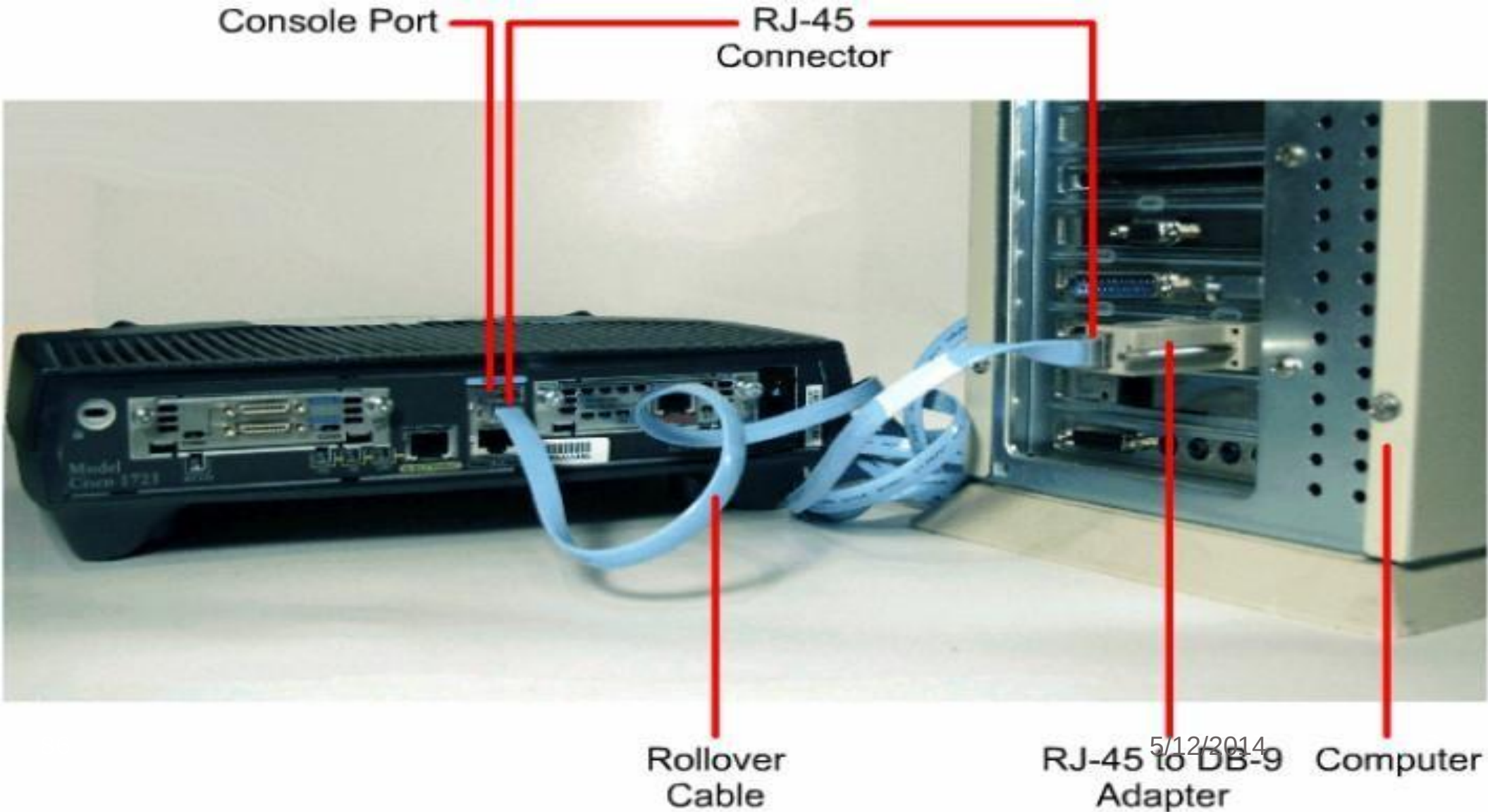
1. **Directly** using the console port
 - ✓ No configuration needed on the router
2. **Telnet** – over the network
 - ✓ Network connection
 - ✓ Network configuration on both ends (IP address, subnet mask, etc.)
 - ✓ Telnet password configured on router
3. **Modem**

Accessing the router directly

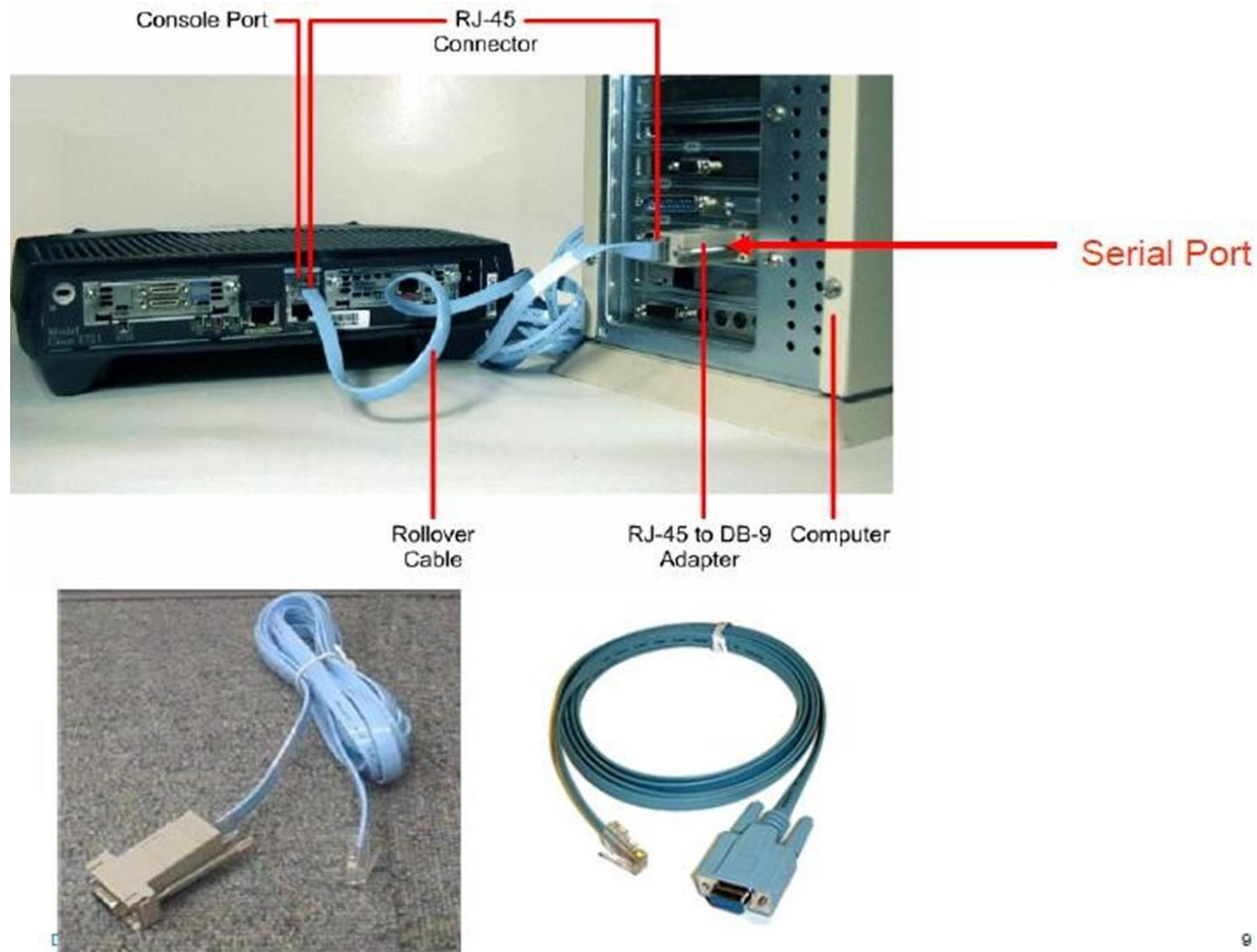
- The console port on the router allows you to connect a “dumb” terminal or a computer with terminal emulator software directly.
- No IP addressing or other configurations needed on the router.
- Used for:
 - ✓ Configuring a device that has never been configured.
 - ✓ Configuring a device directly, because you are physically located where the device is.
 - ✓ Troubleshooting a device that you can no longer access remotely.
 - ✓ Performing password recovery.



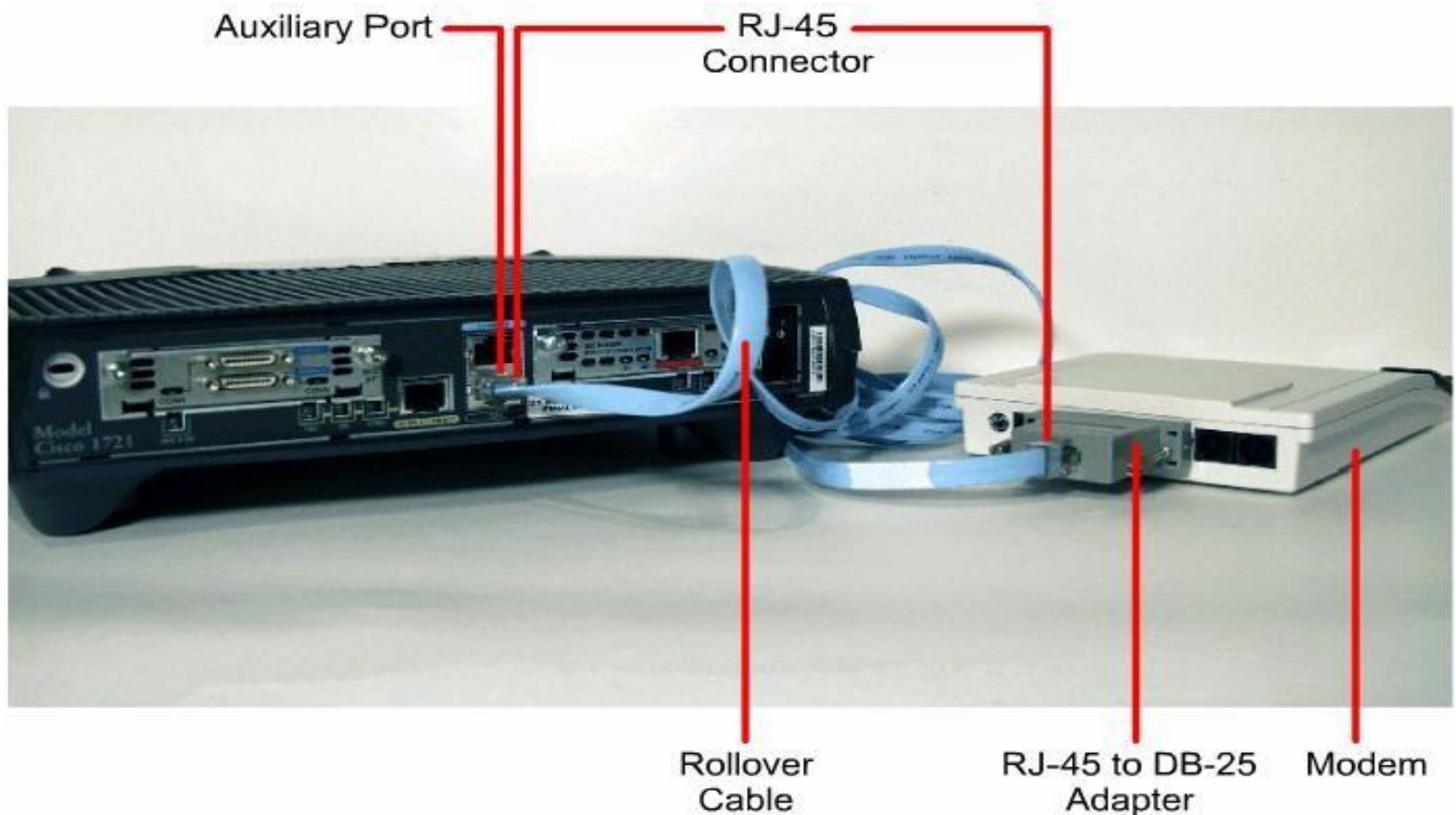
Management port – Console Port connections



Management port – Console Port connections

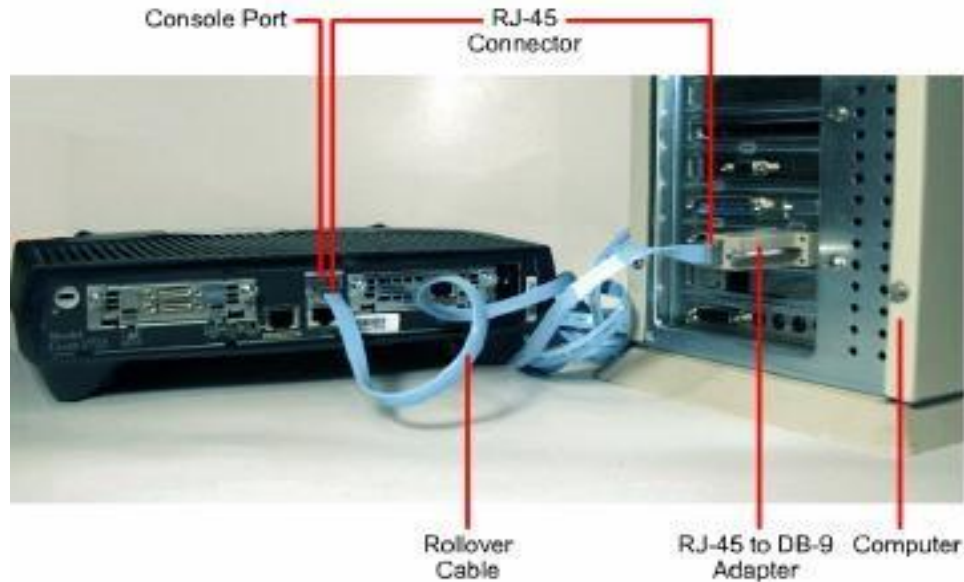


Management port –A U X port connections

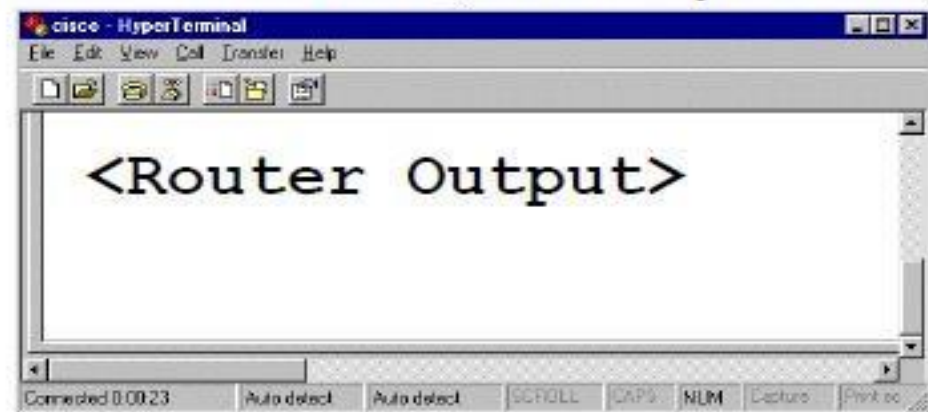


This is for connecting using Modem

Connecting console interfaces

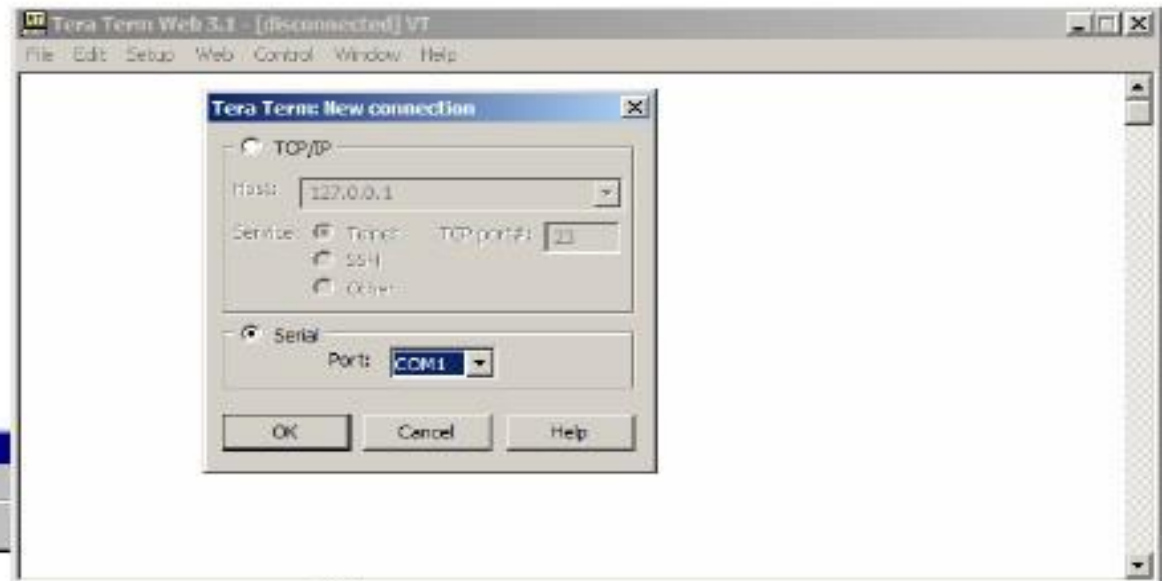


When connected using the console interface, the computer is acting as a “dumb terminal”.

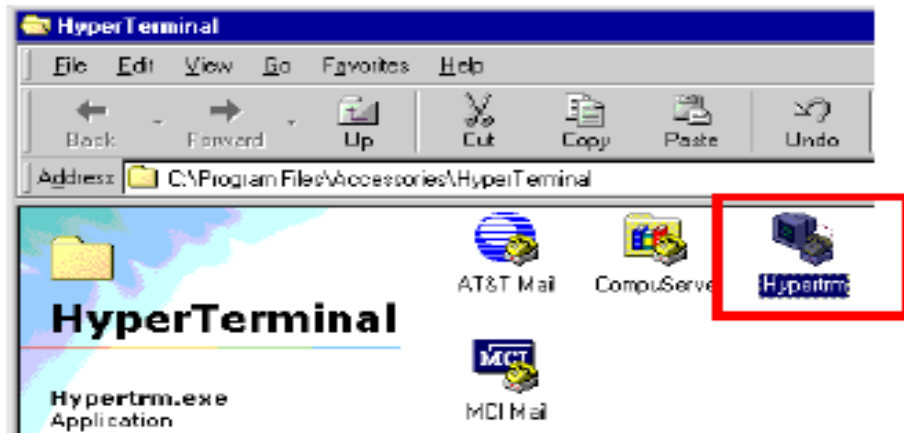


Terminal Emulators

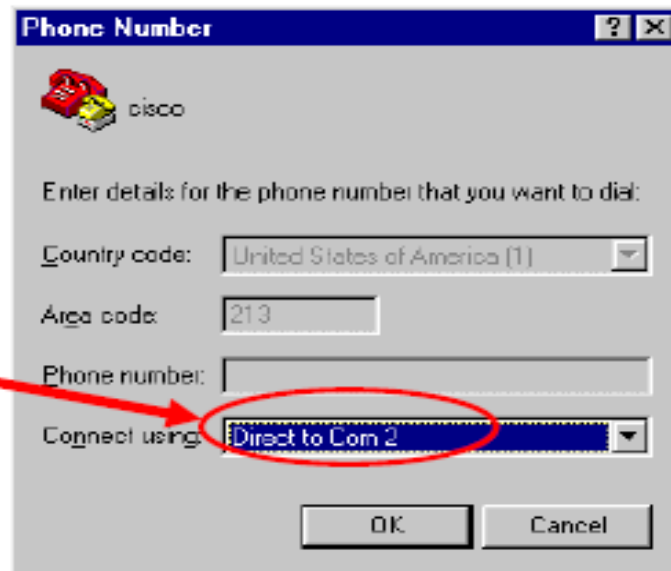
- HyperTerminal: Comes with Windows
- Teraterm: Free download (recommended)
- Settings:
 - 9600 Baud
 - 8 Data Bits
 - 1 Stop Bit
 - No Parity



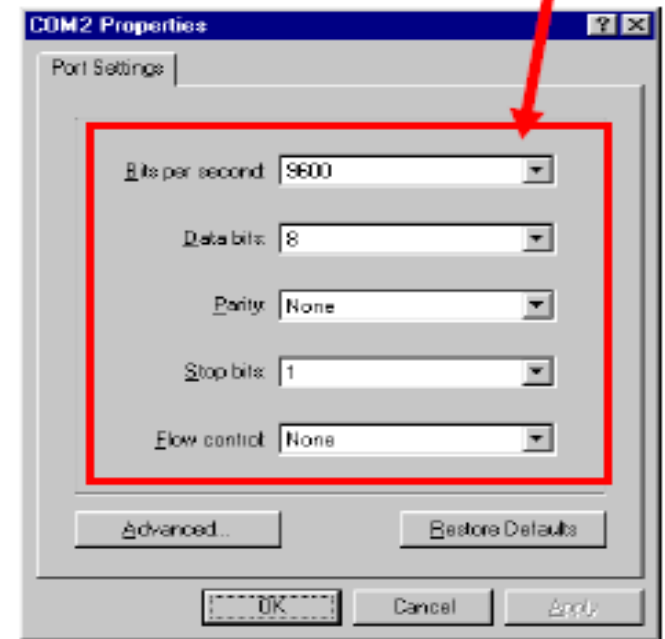
Using HyperTerminal



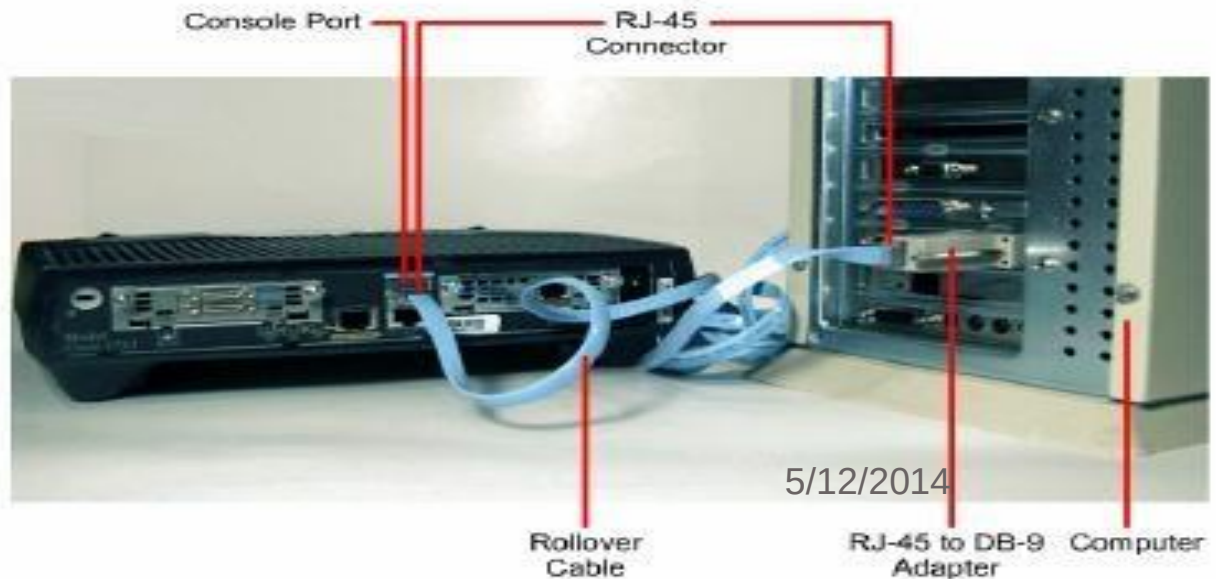
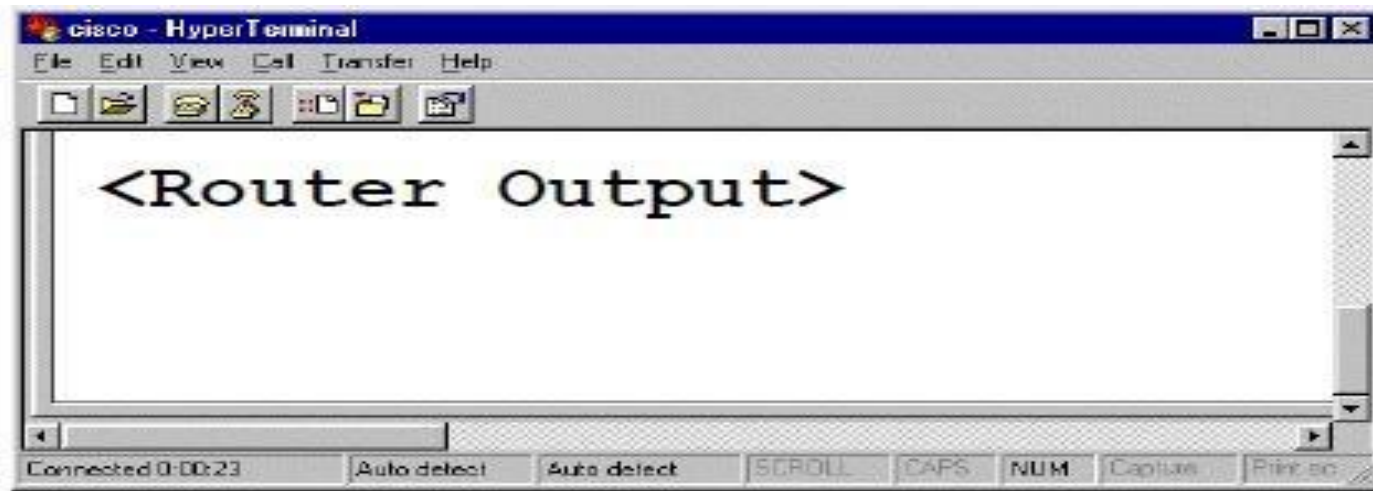
Settings



Match
serial port
on PC



Connecting console interfaces



Connecting LAN interfaces

- A router is a host that communicates with the LAN via a hub or a switch using a **straight through cable**
- The router is connected to the LAN using an Ethernet or Fast Ethernet interface
- A 10/100BaseTX router interface requires UTP, Cat 5 or better cable

Connecting LAN interfaces

Ethernet 10BASE-T Port (RJ-45)

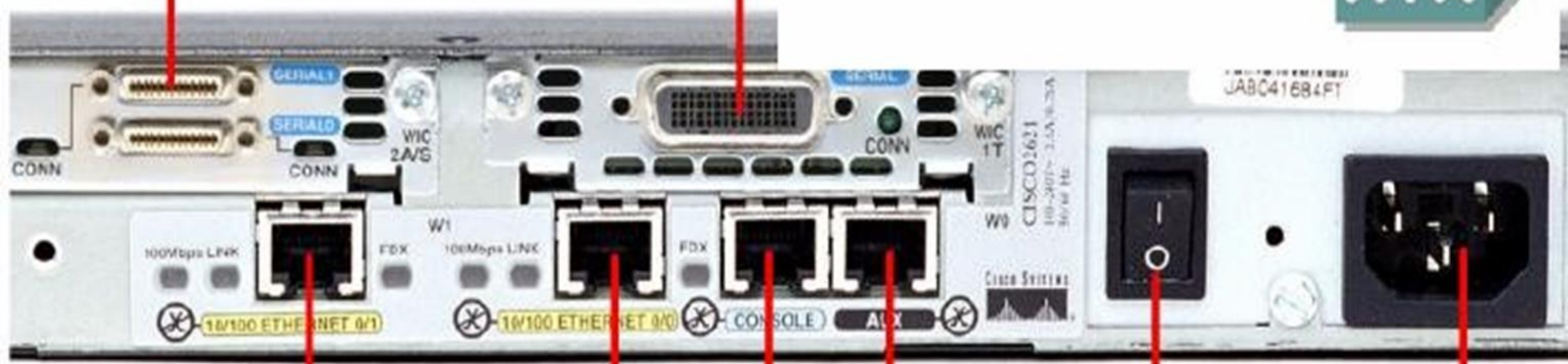


10BASE-T Cable

Ethernet Hub



Serial Ports



FastEthernet Ports

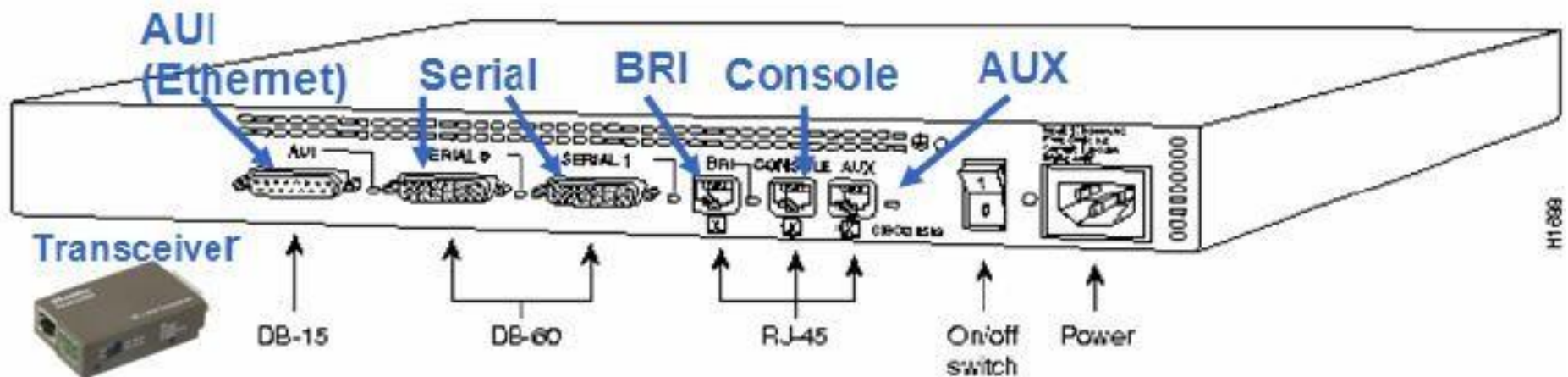
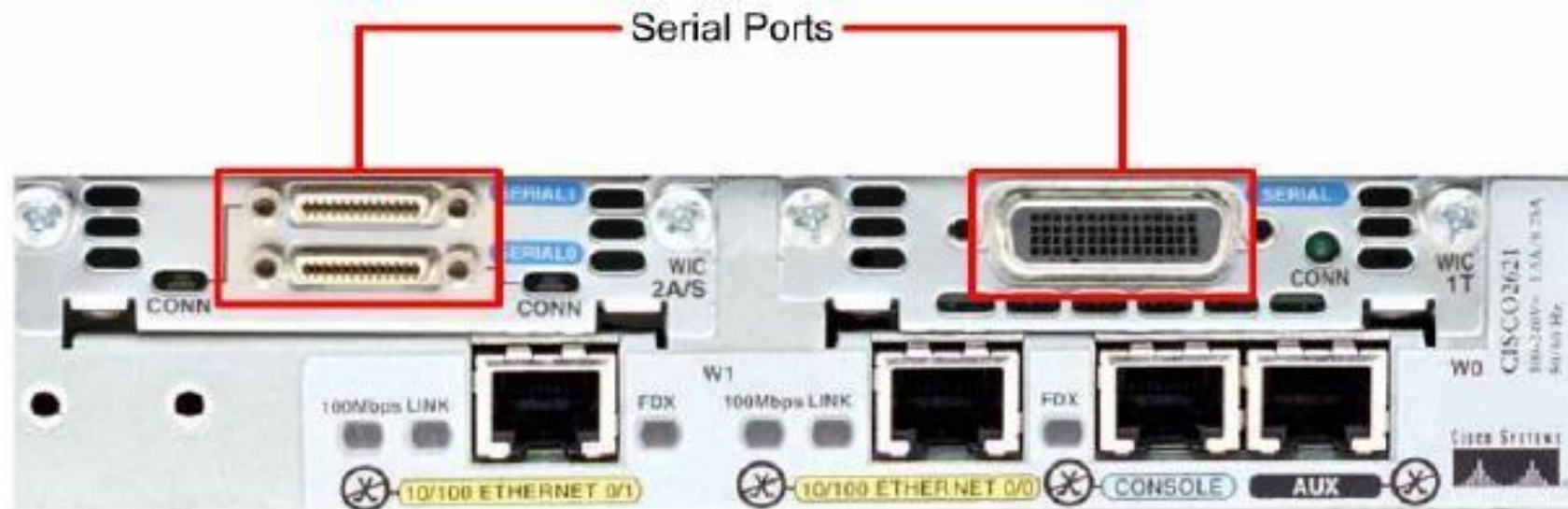
Console Port

Auxiliary Port

Power Switch

Power Cord Connection

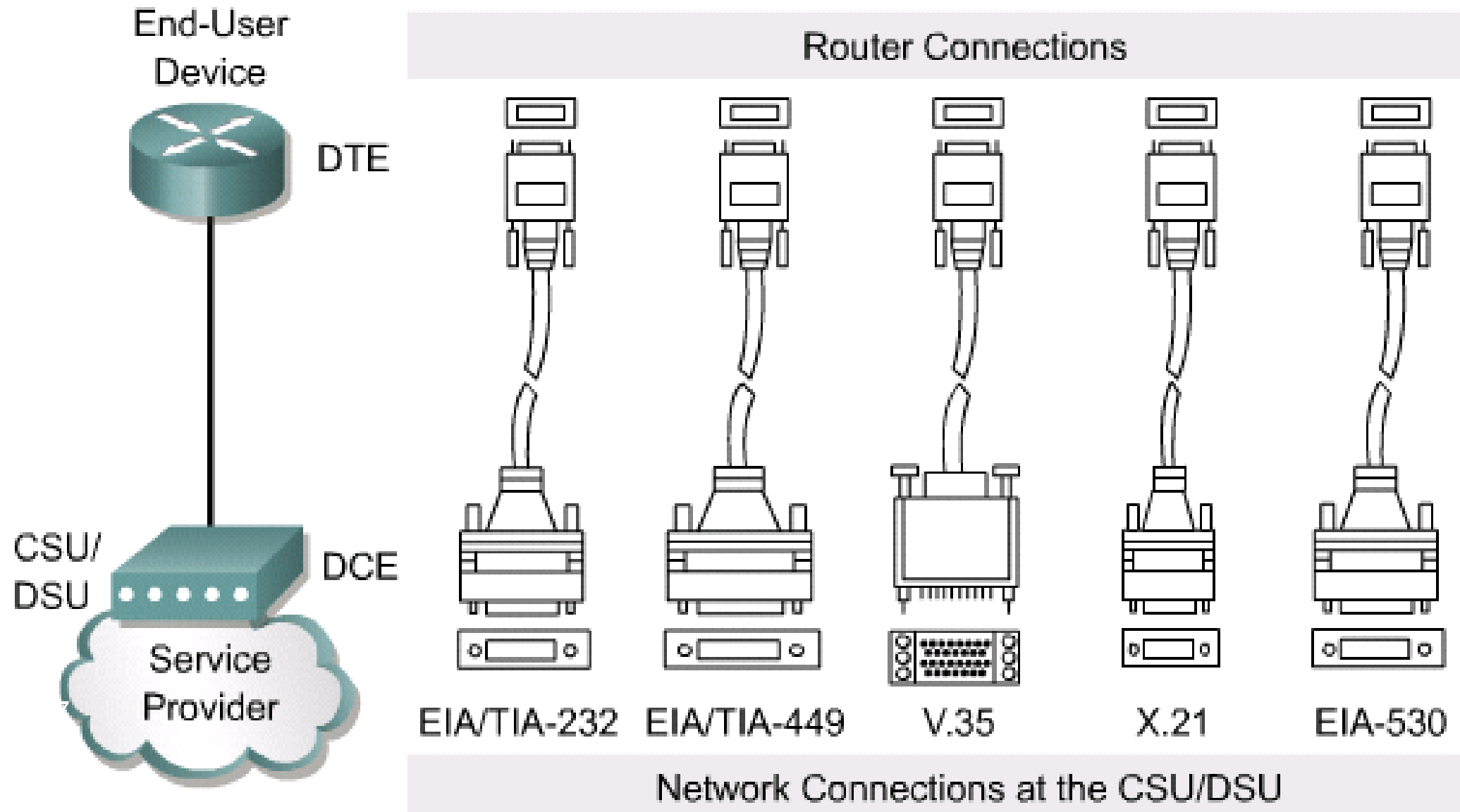
Connecting WAN interfaces



Connecting WAN interfaces

- The customer premises equipment (CPE) is often a router and is the data terminal equipment (DTE)
- The DTE is connected to the service provider using a data circuit-terminating equipment (DCE) device, commonly a modem or channel service unit/data service unit (CSU/DSU).
- The DCE is used to convert the data from the DTE into a form acceptable to the WAN service provider.

Connecting WAN interfaces





NEXT LECTURE
ROUTER FUNDAMENTALS

Homework

- Read more about WANs Technology's
- Read Chapter 1 Introduction to WANs